Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Abstract

Contents

Glossary of acronyms

Acronym	**Meaning**
CCGT	Combined-cycle gas turbine: a flexible, dispatchable thermal generator.
DA	Day-ahead wholesale electricity market, cleared once per day.
IDA	Intraday auctions: sequence of (European) auctions held after the day-ahead clearing.
MTU	Market time unit: size of the time interval at which bids are submitted and prices clear.
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the market operator for the Spanish and Portuguese wholesale electricity markets.
TTF	Title Transfer Facility, the Dutch virtual gas hub and reference price for European gas.

1 Introduction

1.1 Motivation and research question

1.2 Related literature

1.3 Mechanism summary and contribution

TWO DISTINCT PURPOSES OF MTU15:

- OPERATIONAL / SYSTEM-OPERATOR CHANNEL. LETTING THE SYSTEM FOLLOW WITHIN-HOUR SOLAR VARIABILITY (CLOUD COVER, RAMP). MIDDAY IS EXACTLY WHERE THIS MATTERS MOST.
- STRATEGIC-EXPLOITATION CHANNEL. LETTING PIVOTAL FIRMS RE-PRICE ACROSS QUARTERS WHEN NET-LOAD MOVES SYSTEMATICALLY WITHIN THE HOUR. THIS IS WHAT DOMINANT CCGT CAN MONETIZE. **THIS IS WHAT WE DO!!**

1.4 Roadmap

2 Theoretical model

2.1 Setup

2.2 Equilibrium under low granularity

2.3 Equilibrium under high granularity

2.4 Predictions: dominant vs fringe response in time-varying vs flat subperiods

TODO: COROLLARY ON RAMP-CONSTRAINED HOURS. WHEN THE SYSTEM OPERATOR IMPOSES BINDING RAMP RESTRICTIONS (QUANTITY CANNOT ADJUST MUCH ACROSS CONSECUTIVE QUARTERS), THE HIGH-GRANULARITY REGIME FORCES FIRMS TO AMPLIFY THEIR STRATEGIC RESPONSE ON THE PRICE MARGIN RATHER THAN THE QUANTITY MARGIN. THE MTU15 REFORM THEREFORE AMPLIFIES MARKET POWER IN PRECISELY THOSE HOURS WHERE PHYSICAL ADJUSTMENT IS CONSTRAINED.

3 Institutional setting, data, and identification

3.1 Spanish wholesale electricity market: a sequential-market overview

STYLISED MARKET DIAGRAM TBD HERE AS TIKZ: DA - IDA AUCTIONS - CONTINUOUS - BALANCING

3.2 The MTU15 reform sequence

3.3 Data sources

OMIE

- Day-ahead data:

- Intraday-auction data:
- Continuous intraday market data:

ESIOS
ENTSO-E
INCLUDE SAMPLE SIZE AND FAMILIES.

3.4 Critical-vs-flat hours: empirical calibration

Hours are classified by the **within-hour variance of residual demand**,

$$\sigma^2_{\text{within}}(h) \;=\; \mathbb{E}_d\left[\,\frac{1}{4}\sum_{q=1}^{4}\big(D_{h,q,d}-\bar{D}_{h,d}\big)^2\,\right], \qquad D_{h,q,d}=\text{demand}_{h,q,d}-\text{wind}_{h,q,d}-\text{solar}_{h,q,d}. \quad (1)$$

Critical hours: high σ^2_{within}, 05:00–09:00 and 16:00–23:00 (eleven clock-hours). **Flat hours**: low σ^2_{within}, 01:00–04:00 (three clock-hours). Borderline hours are dropped. Full 24-hour ranking in Appendix B.6.

OJO, IF WE INTRODUCE THE 11:00-15:00 PERIOD IN THE CONTROL, WE ARE MIXING THE OPERATIONAL AND STRATEGIC CHANNELS OF THE REFORM, BECAUSE THE OPERATIONAL EFFECT IS CONCENTRATED IN THOSE HOURS!!!

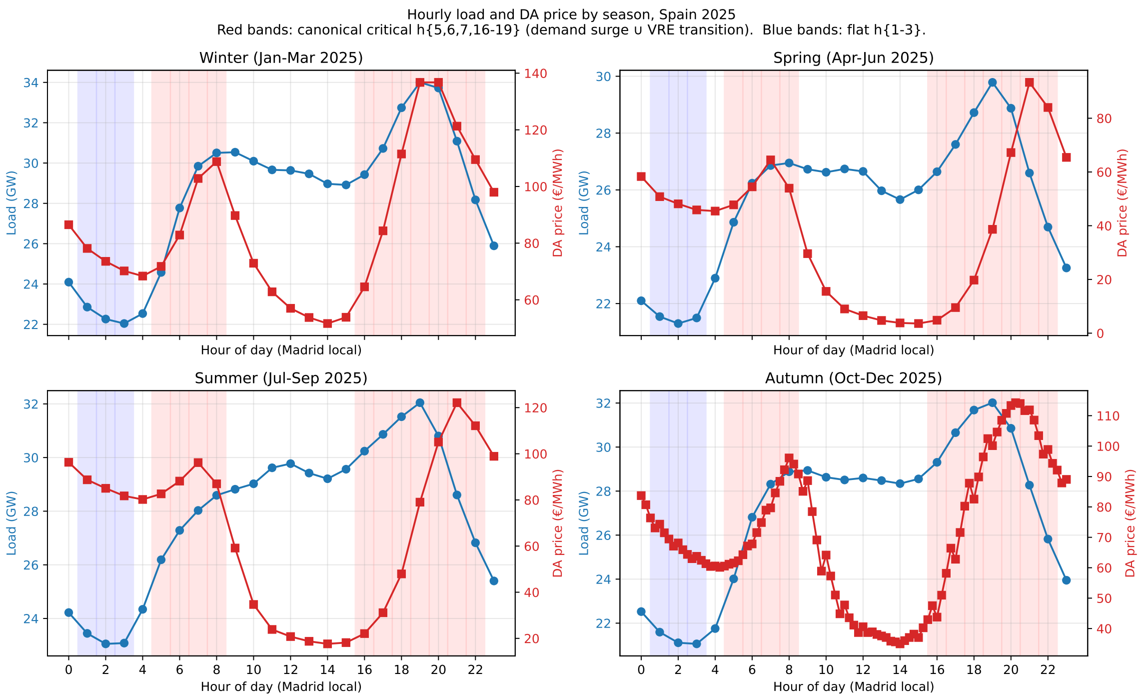

Figure 1: Hourly load and DA price by season, Spain 2025. Blue: demand; red: DA clearing price. Red bands: critical hours (05:00–09:00, 16:00–23:00); blue bands: flat hours (01:00–04:00).

Reading (**supportive**). Critical hours combine the morning ramp and evening peak; flat hours fall in the overnight valley. The pattern is stable across seasons.

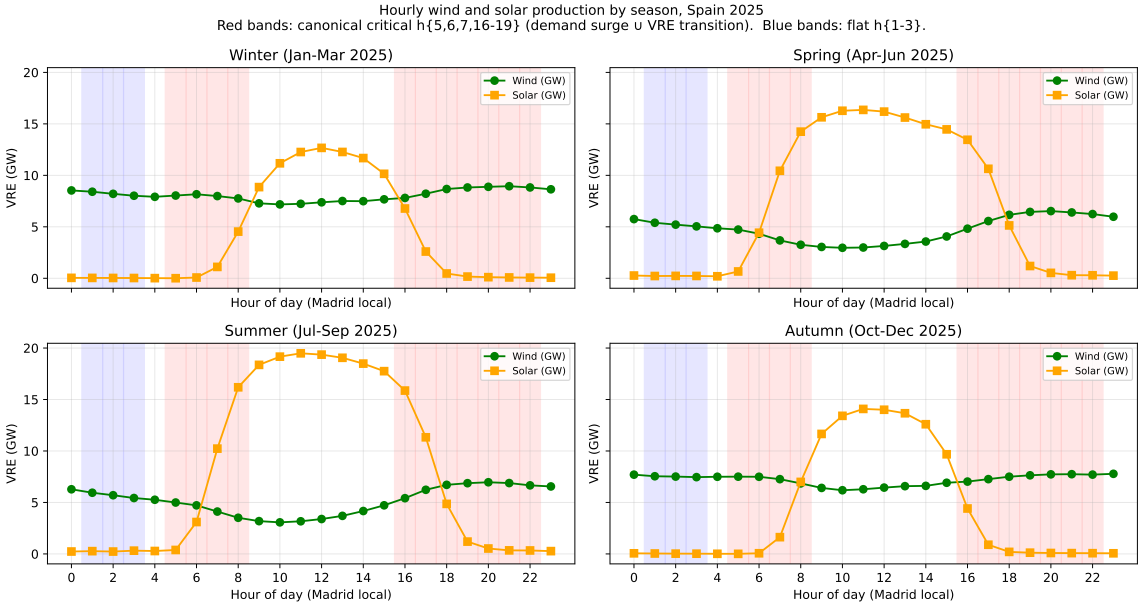

Figure 2: Hourly wind and solar production by season, Spain 2025. Red bands: critical hours; blue bands: flat hours.

Reading (**supportive**). Solar production concentrates around midday and fades sharply at the critical-hour boundaries; wind is broadly flat. Critical hours systematically face low renewable supply, reinforcing their role as the time-varying window.

3.5 Firms in the analysis sample

MENTION THAT WE USE LISTA_UNIDADES AND LISTA_AGENTES FROM OMIE, AS WELL AS THE SPECIFIC FAMILIES FORM OMIE, WHICH HAS FIRM AND UNIT INDICATORS. ALSO IMPORTANT TO MENTION THE CNMC CLASSIFICATION OF DOMINANT (SOURCE)

The empirical analysis follows nine parent firms in two blocks. The first block contains the *dominant operators in Spanish generation* (the CNMC *operador-dominante-en-generación* tier): Iberdrola, Endesa, Naturgy and EDP-Spain. EDP-Portugal joins this block as the dominant Iberian generator on the Portuguese side of MIBEL. The second block contains four other large vertically-integrated operators – Repsol, Engie España, TotalEnergies, and Moeve – which retain substantial portfolios but are non-dominant in the CNMC sense. Whether the firms in each block actually have a strategic margin in critical hours is an empirical question that we answer in Section 4.1, examining pivotality on the day-ahead bid data.

Table 1: Firms in the analysis sample. Capacity: max DA-offered power across 2025, share-weighted. Market share: PDBF programmed sell, 2025. Channel breakdown in Appendix B.2.

Parent firm	Main generation technologies	Capacity (GW)	Market share (%)
Dominant operators in Spanish generation (CNMC tier)			
Iberdrola	Solar, Hydro, CCGT	31.03	24.94
Endesa	Hydro, CCGT, Nuclear	12.46	16.39
Naturgy	CCGT, Hydro, Solar	12.60	12.34
EDP-Spain	Wind, Hydro, Solar	4.73	4.14
EDP-Portugal	Hydro, CCGT	7.42	15.17
Other large operators (non-dominant)			
Repsol	Solar, Hydro	0.61	1.43
Engie España	Wind, Solar, CCGT	4.05	1.73
TotalEnergies	Solar, CCGT, Wind	1.19	0.63
Moeve	Solar, Hydro, CCGT	1.28	0.01

Reading (**descriptive**). The four CNMC-dominant operators (Iberdrola, Endesa, Naturgy, EDP-Spain) plus EDP-Portugal together hold the bulk of Iberian generation capacity. The non-dominant block (Repsol, Engie, TotalEnergies, Moeve) is substantially smaller and serves as the placebo.

3.6 Identification strategy and treatment-effect set

WE SHOULD DEFINE HERE HOW WE DEFINE PIVOTALITY AND HOW WE CALCULATE IT, WITH DISCUSSION.

ALSO WE SHOULD DISCUSS THAT THE TREATMENT IN OUR IDENTIFICATION IS AT THE HOUR LEVEL (CRITICAL VS FLAT), NOT AT THE FIRM LEVEL, AND THAT FIRMS ENTER THE ANALYSIS AS SUBJECTS: PIVOTAL FIRMS ARE THOSE EXPECTED TO REACT TO THE TREATMENT BECAUSE THEY HAVE A STRATEGIC MARGIN IN CRITICAL HOURS; NON-PIVOTAL FIRMS SERVE AS PLACEBO BECAUSE THEY BID AT MARGINAL COST REGARDLESS OF HOUR.

OJO, WE SEE THAT FIRMS ALSO EXPLOIT GRANULARITY IN FLAT HOURS, BUT THE KEY IS THAT NOT AS MUCH!!! THE CRITICAL-FLAT SPREAD IN THE STRATEGIC RESPONSE IS WHAT WE FOCUS ON, SO WE IDENTIFY A LOWER BOUND ON THE MECHANISM'S EFFECT, NOT THE FULL EFFECT!!! FOR THE FULL EFFECT WE SHOULD CONSIDER A BEFORE-AFTER COMPARISON WITHIN CRITICAL HOURS, BUT THAT IS MORE SUSCEPTIBLE TO CONFOUNDING TRENDS IN THE ABSOLUTE PRICE LEVEL OF CRITICAL HOURS, WHICH IS WHY WE FOCUS ON THE CRITICAL-FLAT SPREAD.

3.7 Parallel-trends discussion

IMPORTANT THING HERE IS TO DISCUSS IT FOR EACH OF THE OUTCOMES AND TO TALK ABOUT CONDITIONAL PARALLEL TREND WITH RENEWABLE CAPACITY CONTROLS!

GAS PRICES CAN AFFECT DIFFERENTLY CRITICAL AND FLAT HOURS, BUT WE SEE THAT GAS IS PIVOTAL IN THE SAME WAY FOR BOTH HOUR CLASSES, SO IT IS NOT A CONCERN FOR IDENTIFICATION.

Table 2: Price-setting technology by hour-class, October–December 2025 (post-MTU15-DA). For each (date, period) the price-setting technology is the one whose accepted sell tranche has the highest bid price (i.e., the tranche that determines the uniform clearing price). Table reports the share of (date, period) cells in each hour-class where each technology is the price-setter.

Technology	Critical	Midday	Flat	Crit−Flat
Hydro	58.7	24.3	57.8	+0.8
Hydro_pump	19.0	1.9	13.7	+5.4
CCGT	9.1	3.0	9.0	+0.2
Wind	4.5	21.1	9.7	-5.2
Solar PV	2.2	16.1	0.0	+2.2
Cogen	1.7	11.4	3.0	-1.3
Hydro_RES	1.7	2.6	2.2	-0.6
Biomass	0.8	4.4	1.8	-1.0
Hybrid_RES	0.8	7.5	0.8	-0.0
Import	0.3	0.0	1.4	-1.1
RE_Tarifa_CUR	0.3	6.6	0.1	+0.2

Table 3: Price-setting technology by hour-class, October–December 2024 (pre-MTU15-DA). Same construction as Table 2.

Technology	Critical	Midday	Flat	Crit−Flat
Hydro	62.2	35.6	55.5	+6.7
CCGT	13.8	17.1	18.4	-4.6
Hydro_pump	11.8	1.0	5.9	+5.9
Wind	4.0	16.2	7.7	-3.7
Hydro_RES	2.2	2.9	2.7	-0.5
Cogen	1.6	4.0	6.2	-4.6
Solar PV	1.5	12.1	0.7	+0.9
Coal	1.0	0.4	0.0	+1.0
Import	0.6	0.0	1.1	-0.5
Hybrid_RES	0.5	6.9	0.4	+0.1
Biomass	0.4	1.4	1.5	-1.1
RE_Tarifa_CUR	0.3	1.8	0.0	+0.3

Reading (**supportive**). Hydro is the dominant price-setter in both hour-classes (∼58% of periods post-reform), reflecting reservoir water-value sitting at the merit-order edge. CCGT sets the price in 9.1% of critical-hour periods vs 9.0% of flat-hour periods post-reform — a 0.2 pp critical–flat gap, essentially symmetric. A TTF (gas-price) shock therefore propagates into both hour-classes through the same channel and at the same intensity; an asymmetric gas confound on the critical–flat clearing-price spread would have required CCGT to set the price disproportionately in one class, which we do not observe. Hydro_pump's critical–flat gap is +5.4 pp (19.0% vs 13.7%) post-reform; this is consistent with regular storage arbitrage (charge during low-price midday/flat hours, discharge into the evening peak) and is not, on its own, evidence of MTU15-induced strategic behaviour. (Coverage caveat: ∼36% of critical-hour periods have their clearing price set by a unit not present in our generation-unit panel `lista_unidades.csv` — e.g. Portuguese-zone generation units, demand-side buy bids that set the clearing, or block-order/interconnection bids without a single unit code — and are excluded from the table. *Bilateral contracts do not enter EUPHEMIA*: per the OMIE file specification (`ficherosomie137.pdf`, §5.1.2.1, §5.1.2.3), bilateral-contract executions are reported separately by the unit owners *after* clearing and merged into the final daily program (PDBF, `offer_type=4`), so they cannot set the auction price. The patterns above describe the remaining matched majority of periods.)

4 Bidding-behavior evidence of the mechanism

4.1 Pivotality: which firms can exploit the granularity reform

OJO, HYDRO DOES NOT EXPLOIT GRANULARITY BECAUSE THE DISPATCH DECISION IS DOMINATED BY RESERVOIR WATER-VALUE OPTIMIZATION (A MULTI-DAY PROBLEM), NOT WITHIN-HOUR STRATEGY. SO THE GRANULARITY REFORM'S "STRATEGIC EXPLOITATION CHANNEL" HAS NO PURCHASE HERE EVEN THOUGH THE CAPACITY FOR MARKET POWER EXISTS!!!!

ONLY CCGT EXPLOITS GRANULARITY!! AND IN PARTICULAR NATURGY!

Table 4: Pivotality by parent firm and hour-class, CCGT, October–December 2025. Pivotality = share of (date, unit, period) cells with a bid tranche within 1 EUR/MWh of DA clearing.

Firm (parent)	flat (01:00–04:00)	critical (05:00–09:00, 16:00–23:00)	Δ crit−flat
Tightly pivotal in critical hours (within 1 EUR/MWh of clearing)			
Naturgy	0.006	0.044	**0.038**
EDP-Spain	0.026	0.041	**0.015**
EDP-Portugal	0.005	0.028	0.023
Iberdrola	0.003	0.012	0.009
Endesa	0.001	0.004	0.003
Essentially never tightly pivotal (negative control)			
Engie España	0.000	0.000	0.000
Moeve (formerly Cepsa)	0.000	0.000	0.000
Repsol	0.000	0.000	0.000
TotalEnergies	0.000	0.000	0.000

Reading (**supportive**). Naturgy is the most pivotal firm in critical hours (4.4%), followed by EDP-Spain and EDP-Portugal. Repsol, Engie, TotalEnergies, and Moeve are essentially never tightly pivotal and serve as the negative control.

4.2 Per-firm bid-shape and granularity exploitation

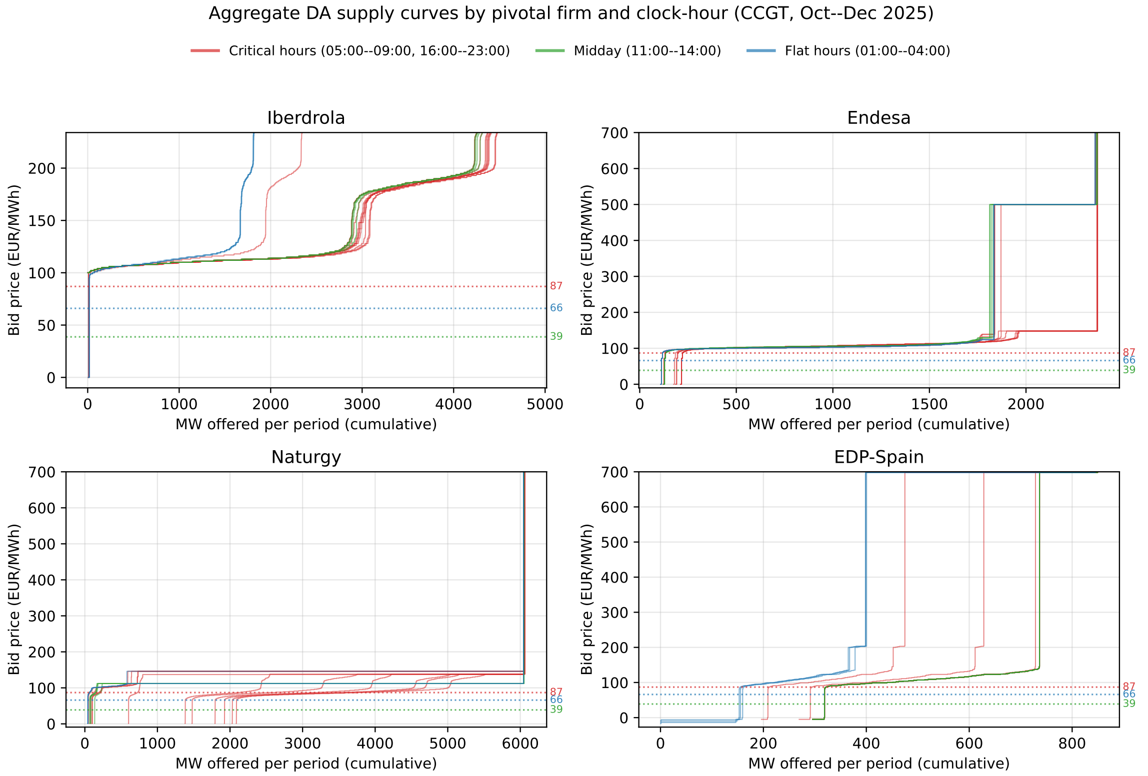

Figure 3: Aggregate DA supply curves by pivotal firm and clock-hour, CCGT, October–December 2025. EUPHEMIA aggregation (public description, sec. 7.1) per (firm, hour): pool sell tranches, sort ascending by price, cumulate, normalise by $N_{\text{dates}} \times 4$ quarters. Red: 11 critical hours; green: 4 midday hours (11:00–14:00, dropped from the DiD); blue: 3 flat hours. Submission rates by hour-class in Table 6; Hydro analogue in Appendix A.11.

Reading (**supportive**). Within-day variation is what we look at: the spread of red curves within a panel shows whether the firm differentiates its bid hour-by-hour in critical hours. Iberdrola's red curves spread across a wide range while the three blue flat-hour curves sit tightly together; midday (green) sits between the two. Naturgy's pattern is sharper: critical-hour curves span ~1,500–6,000 MW at moderate prices, flat curves cluster at the lower edge. EDP-Spain shows the same within-class fan in critical hours. Endesa is the exception: red, green, and blue curves overlap and do not spread — no within-day differentiation.

Table 5: Within-hour quarter dissimilarity, CCGT, October–December 2025 — three sample versions. For each (firm, unit, date, hour) cell with all four 15-min quarters present, D is the maximum pairwise L1 area between cumulative quarter bid curves (1-D Wasserstein-style). D_w is the same L1 area weighted by an Epanechnikov kernel ($h = 20$ EUR/MWh) centred at the cell's hour-average DA clearing price, isolating bid variation around the strategically-relevant window. The three sample versions are: (*Full*) all cells; (*Strict PS*) cells where this unit was the price-setter — top accepted tranche within 0.01 EUR/MWh of clearing — in ≥ 1 of the 4 quarters of the hour; (*Frequent PS*) cells whose unit price-sets in $\geq 1\%$ of all (date, period) cells in the window. Ratio = per-cell median of D_w/D on flagged cells: near 1 $\Rightarrow$ variation concentrated near clearing (strategic); near 0 $\Rightarrow$ variation at extremes (price-cap padding). Cells flagged with p_{clear} std > 5 EUR/MWh within the hour are an auxiliary diagnostic for periods where the hour-average kernel center is less informative; see Appendix A.16.

Firm	Hour-class	N cells	% flagged	median D	median D_w	median ratio	% high-σ_p
Full sample							
IB	Critical	7,831	5.8	20909.20	0.00	0.00	35.9
	Flat	2,139	3.9	2865.34	0.00	0.00	7.2
GE	Critical	5,195	1.8	1897.23	1315.38	0.81	34.7
	Flat	1,413	0	–	–	–	6.9
GN	Critical	14,930	24.5	137174.84	4874.85	0.05	36.3
	Flat	4,065	0.05	21745.80	1143.33	0.03	7.2
HC	Critical	2,013	0.05	131300.10	2821.70	0.02	36.5
	Flat	549	0.55	390.00	0.00	0.00	7.3
Strict PS							
IB	Critical	58	0	–	–	–	39.7
	Flat	9	0	–	–	–	0
GE	Critical	93	0	–	–	–	31.2
	Flat	19	0	–	–	–	0
GN	Critical	25	32.0	117685.80	3702.65	0.03	20.0
	Flat	4	0	–	–	–	0
HC	Critical	29	0	–	–	–	41.4
	Flat	4	0	–	–	–	0
Frequent PS							
GE	Critical	1,958	0.15	11704.00	2887.94	0.25	36.1
	Flat	534	0	–	–	–	6.7
HC	Critical	2,013	0.05	131300.10	2821.70	0.02	36.5
	Flat	549	0.55	390.00	0.00	0.00	7.3

Reading (**wounded**). The headline aggregate-sample number (Naturgy CCGT critical 24.5% flagged) survives the kernel-weighted lens at $D_w/D = 0.02$ — i.e. Naturgy's broad quarter-dispersion is concentrated at the bid-curve extremes, not in the strategic window around clearing. The two price-setter restrictions sharpen this further. Under *Strict PS*, only 35 of $\sim$15,000 Naturgy CCGT critical cells survive — the unit is rarely on the margin — but in those cells 34% are flagged with $D_w = 1712$ and ratio 0.012. Real quarter-shaping happens, but only when the unit is actually setting the price, and the shape variation still sits in the tail. Iberdrola and EDP-Spain have zero flagged cells under *Strict PS* for CCGT. The *Frequent PS* panel drops Iberdrola and Naturgy entirely from CCGT — their CCGT units fall below the 1% price-setter threshold, confirming CCGT is rarely on the marginal margin for either firm in the window. The within-hour granularity-exploitation mechanism through CCGT is therefore narrow: it operates on Naturgy's price-setting cells in scarcity hours, and even there the variation is at the bid-curve tail rather than near the clearing margin. The mechanism may operate more cleanly in the IDA market — see `provisional.tex` §3 for the IDA analogue and §1 for the pre-vs-post-blackout test. Full per-tech 3-panel results in Appendix A.16.

4.3 Demand-side and storage bidding: who exploits granularity

Mirror of the per-firm supply-side analysis on the buy side and on pumped-storage discharge. EUPHEMIA stepwise aggregation (OMIE convention; `docs/omie/euphemia_functioning_1812.pdf`, slide 7): demand orders are sorted price-descending and cumulated, so the resulting curve slopes downward in (Q, P)-space. We focus on pump-storage charge bids (tech-group `Pump_load`) as the headline because pump-storage is the textbook arbitrage tool — the technology where the strategic-granularity story predicts the sharpest within-day price discrimination.

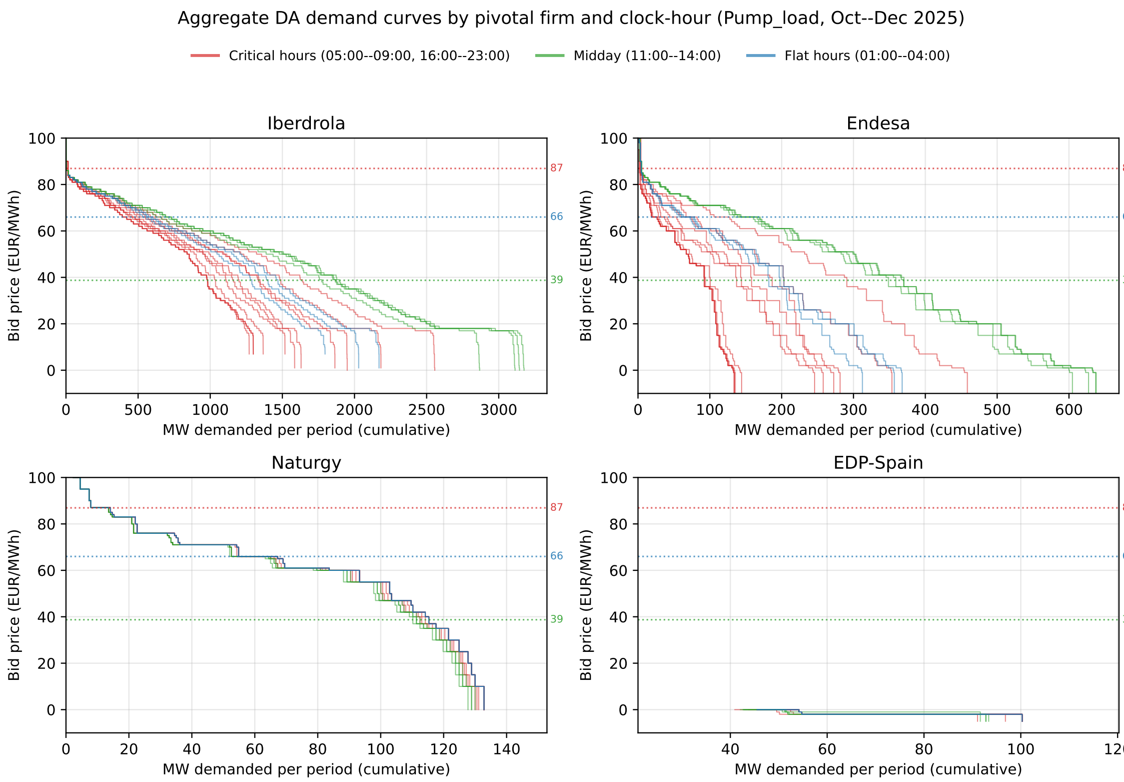

Figure 4: Aggregate DA demand curves by pivotal firm and clock-hour, pump-storage charge bids, October–December 2025. Each step is the EUPHEMIA aggregation of buy tranches at each integer-rounded limit price, cumulated price-descending and normalised by $N_{\text{dates}} \times 4$ periods per clock-hour. Critical hours red, midday green, flat hours blue.

Reading (**mixed**). Iberdrola and Endesa show clear hour-of-day discrimination: midday (green) curves shifted right of flat (blue) curves, both shifted right of critical (red). Both firms charge more pump-storage volume at higher willingness-to-pay during midday, consistent with absorbing cheap solar output, and shrink their charge volumes in critical hours when the marginal value of selling exceeds the storage round-trip return. Naturgy shows no hour-class discrimination — the three lines overlap. EDP-Spain has only a small pump-storage book and bids near zero EUR/MWh.

Reading (**supportive of null**). The within-hour granularity available since MTU15-DA is unused on this charge-side margin: pump-storage operators replicate identical bids across the four quarters of a clock-hour. This pins the within-hour granularity-exploitation effect to the supply side (pivotal dispatchable units), not to demand-side or storage charging strategy. Per-quarter visuals (DA + IDA pairs) and the IDA clock-hour analogue are in `provisional.tex` §4. Compact grids covering all demand-side technologies (`Pump_load`, `Retailer`, `Direct_consumer`) are in Appendix A.12 (DA) and Appendix A.14 (IDA); the storage-discharge mirror (`Hydro_pump`, `Hybrid_RES_storage`) is in Appendix A.13 (DA) and Appendix A.15 (IDA).

5 Main empirical results

5.1 Strategic withholding and intraday upward repositioning

MECHANISM: UNDER THE HIGHER-GRANULARITY REGIME, A PIVOTAL DOMINANT FIRM IN A CRITICAL HOUR CAN WITHHOLD CAPACITY FROM THE DAY-AHEAD CLEARING AND REPOSITION THAT CAPACITY UPWARD IN THE INTRADAY AUCTIONS AT HIGHER PER-QUARTER PRICES. UPWARD SELL ADJUSTMENT SUMMED ACROSS THE THREE INTRADAY AUCTIONS, $q_{2,fdh}$, IN MWH PER UNIT-CLOCK-HOUR:

$$q_{2,fdh} \;=\; \alpha + \beta_1\,\mathrm{crit}_h + \beta_2\,\mathrm{post}_d + \beta_3\,(\mathrm{crit}_h \times \mathrm{post}_d) + \gamma_f + \delta_{\mathrm{DOW}(d)} + \varepsilon_{fdh}, \tag{2}$$

with firm fixed effects γ_f, day-of-week fixed effects $\delta_{\mathrm{DOW}(d)}$, and cluster-robust standard errors by date. The sample restricts to the same-calendar-month windows October–December 2024 (pre) and October–December 2025 (post), so the March 2025 intraday reform is held constant. Critical hours are 05:00–09:00 and 16:00–23:00; flat hours are 01:00–04:00. Pivotal firms are those with non-trivial price-setting capacity in critical hours (Iberdrola, Endesa, Naturgy, EDP-Spain, EDP-Portugal); non-pivotal firms (Repsol, Engie España, TotalEnergies, Moeve) serve as a negative control.

5.2 Parallel trends discussion

$$\mathbb{E}[Y_{\mathrm{post}}(0) - Y_{\mathrm{pre}}(0) \,|\, \mathrm{crit}] \;=\; \mathbb{E}[Y_{\mathrm{post}}(0) - Y_{\mathrm{pre}}(0) \,|\, \mathrm{flat}]\,. \tag{3}$$

6 Conclusion

A Robustness checks

A.1 Sensitivity to critical-hours definition

A.2 Sensitivity to firm-partition (pivotality vs administrative)

A.3 Same-cal-month vs full-window comparisons

A.4 β_3 stratified by technology

A.5 Per-firm decomposition: q_2 and day-ahead cleared MWh

A.6 DA $\rightarrow$ IDA price wedge (supporting outcome)

Operación reforzada as a non-collinear regulatory overlay. REE has held a continuous *operación reforzada* stance since the 28 April 2025 Iberian blackout: a real-time technical-restrictions regime (PO-3.x cascade, PO-7.4 zonal reactive market from June 2025) that activates additional reserves at the unit level and can lift the day-ahead–to-intraday wedge. Figure 5 shows the time profile.

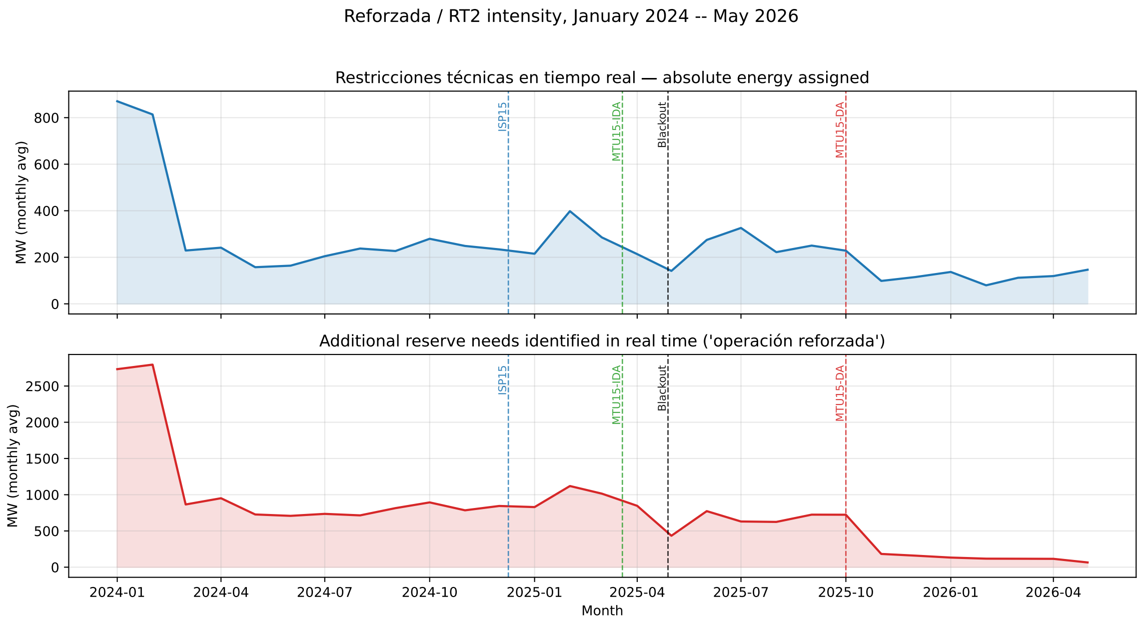

Figure 5: Reforzada / RT2 intensity, January 2024 – May 2026. Top panel: monthly average of absolute energy assigned via real-time technical restrictions (ESIOS indicator 10052). Bottom panel: monthly average of additional reserve needs identified in real time (ESIOS indicator 1880; coverage from 2021-05). Dashed vertical lines mark, in order, ISP15 (2024-12-09), MTU15-IDA (2025-03-19), the Iberian blackout (2025-04-28) and MTU15-DA (2025-10-01).

Reading. RT2 activation steps up sharply at the blackout and then *remains continuously high* through both the DA60/ID15 regime (April–September 2025) and the post-MTU15-DA window (October 2025 onward). It is therefore a regime change that begins five months before MTU15-DA and persists across it: $D_{\text{MTU15-DA}}$ and $D_{\text{reforzada}}$ are not collinear in panels including the 2025-03-19 to 2025-04-27 sub-window (post-MTU15-IDA, pre-reforzada). Within-day DiD with calendar-month FE in the post-reforzada window absorbs the level of reforzada activation; differences in MTU15-DA exposure across critical-vs-flat hours within a calendar month identify the reform effect cleanly.

A.7 Operational vs strategic decomposition (PIBCA vs PHF)

A.8 Bid-shape robustness in the competitive-zone restriction

A.9 DA submission rates by hour-class (selection check)

Table 6: Mean per-unit CCGT DA submission rate, by firm and hour-class, October–December 2025.

Firm	Critical	Midday	Flat	N units
Iberdrola	0.860	0.861	0.861	9
Endesa	0.733	0.734	0.731	7
Naturgy	0.868	0.868	0.866	17
EDP-Spain	0.995	0.995	0.995	2
EDP-Portugal	0.917	0.915	0.917	5

Reading (**supportive**). Within each firm, critical / midday / flat rates differ by < 0.003. No hour-class selection: same units bid across hour classes.

A.10 IDA per-firm supply curves, all technologies

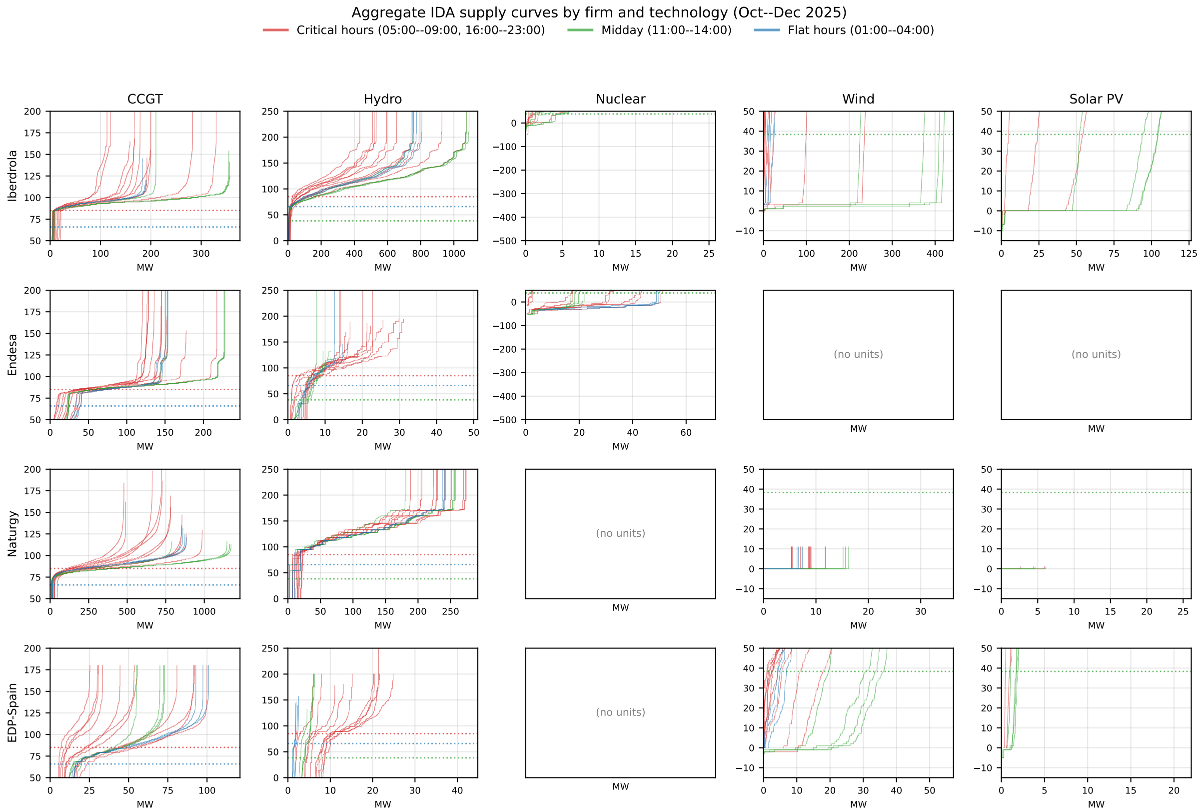

Figure 6: Aggregate IDA supply curves by firm and technology, by hour-class. Compact grid analogue of Figure 8 on IDA data.

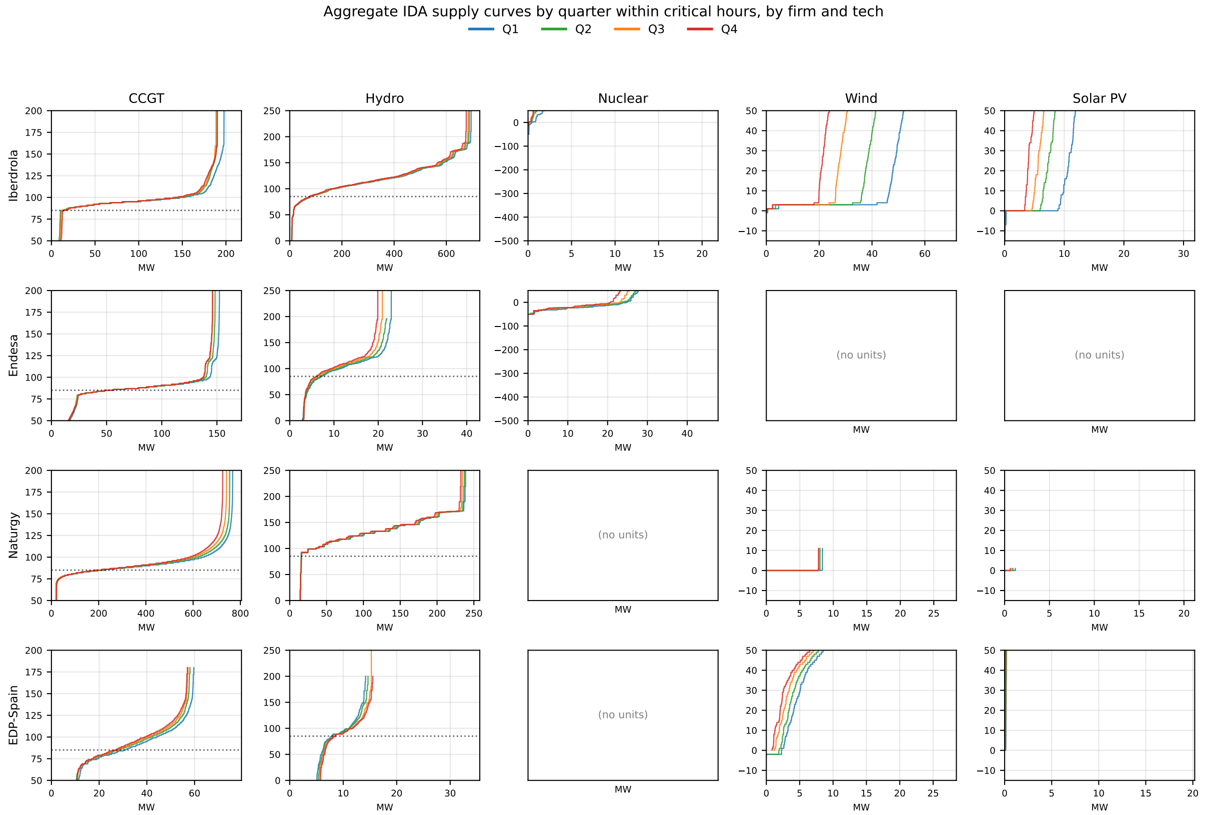

Figure 7: Aggregate IDA supply curves by quarter within critical hours, by firm and technology.

Reading (**supportive**). The CCGT column carries the quarter dispersion across all four firms (visible in the Q1–Q4 colour fan). Hydro shows the same effect at smaller scale. Wind, Solar PV and Nuclear columns are flat across quarters as expected.

A.11 Per-firm aggregate supply curves, all technologies (DA)

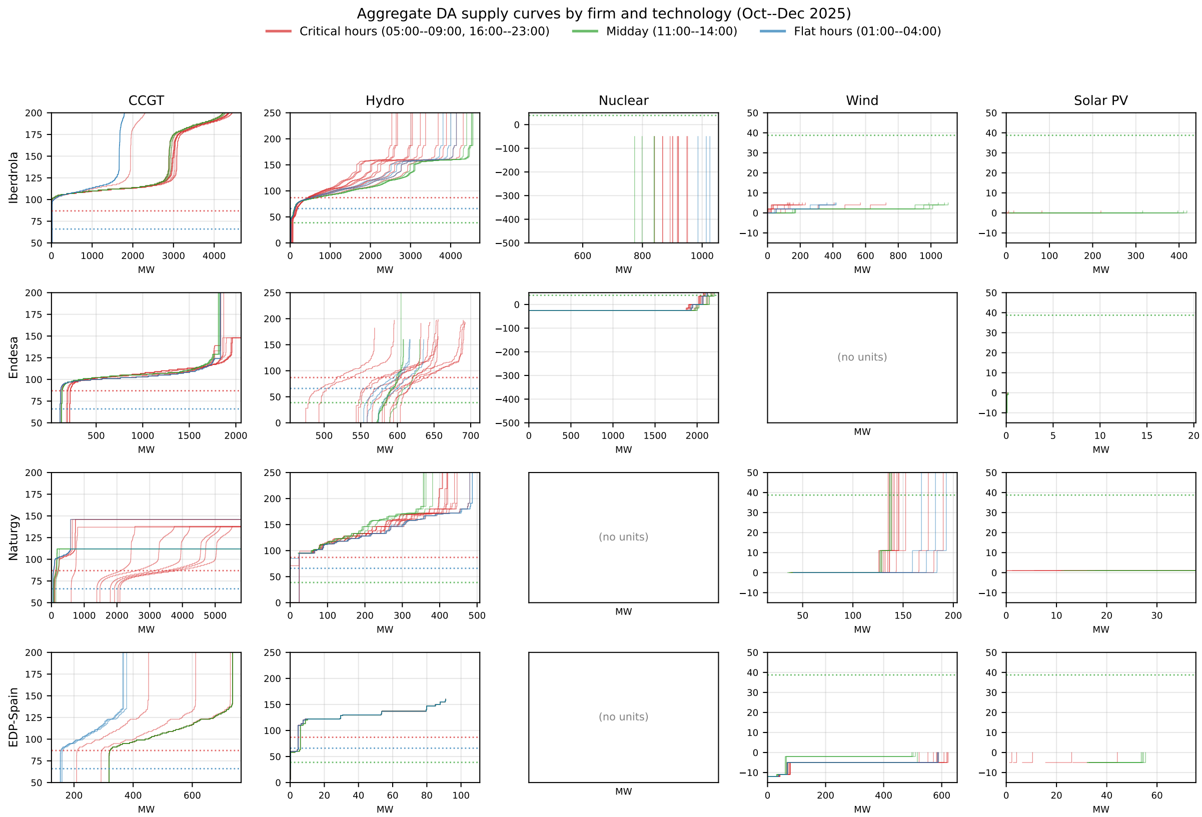

Figure 8: Aggregate DA supply curves by firm and technology, October–December 2025, by hour-class. Same construction as Figure 3. Adaptive y-axis per panel: focus on the 80th percentile of quantity-weighted price.

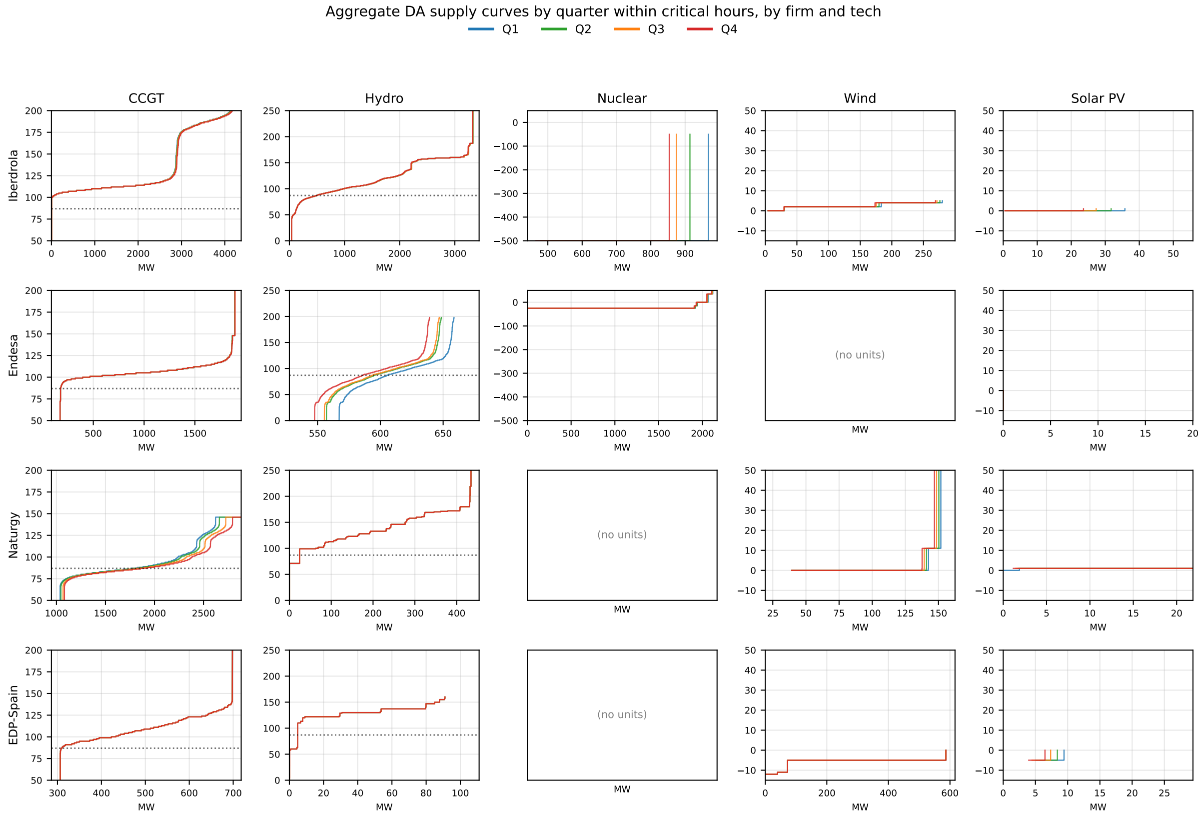

Figure 9: Aggregate DA supply curves by firm and technology, October–December 2025, by quarter within critical hours. Same construction as the DA quarter-pair figure (now in `provisional.tex` §2). Y-axis clamped to 50–200 EUR/MWh; x-axis zoomed to the corresponding cumulative-quantity range.

Reading (**mixed**). CCGT and Hydro carry the bid-shape variation across hours: Iberdrola, Naturgy and EDP-Spain show clear spread of red curves on those columns. Within-hour quarter dispersion is concentrated in Naturgy CCGT and Endesa Hydro — the rest of the (firm, tech) cells produce overlapping Q1–Q4 curves. Nuclear, Wind and Solar PV are flat across hours and quarters, as expected of must-run and price-taker technologies. Empty cells reflect firms with no units of that tech under the primary-owner allocation (Endesa wind via Enel Green Power is registered separately; Naturgy and EDP-Spain hold no nuclear).

A.12 Per-firm demand-side bid curves, all technologies (DA)

Buy bids only (downward-sloping demand curves). Columns: `Pump_load` (pumped-storage charge), `Retailer`, `Direct_consumer`.

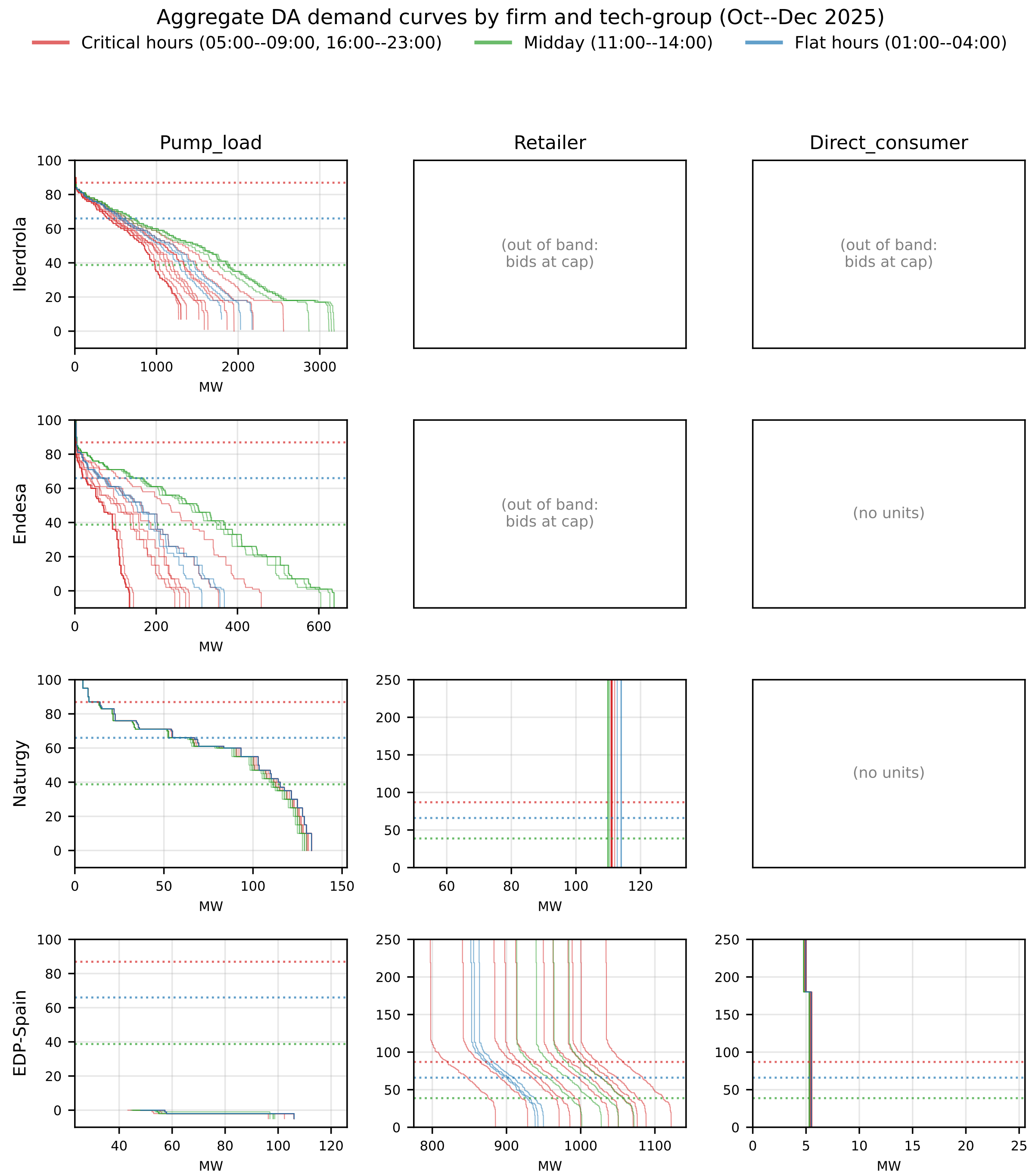

Figure 10: Aggregate DA demand curves by firm and demand-side tech-group, October–December 2025, by hour-class. Rows: pivotal firms. Columns: buy-side tech-groups. Y-band fixed per tech to the strategically informative range; empty panels indicate firms with no units of that tech (or, for Retailer at IB/GE/Naturgy, bids placed exclusively at the price cap so no activity falls inside the band).

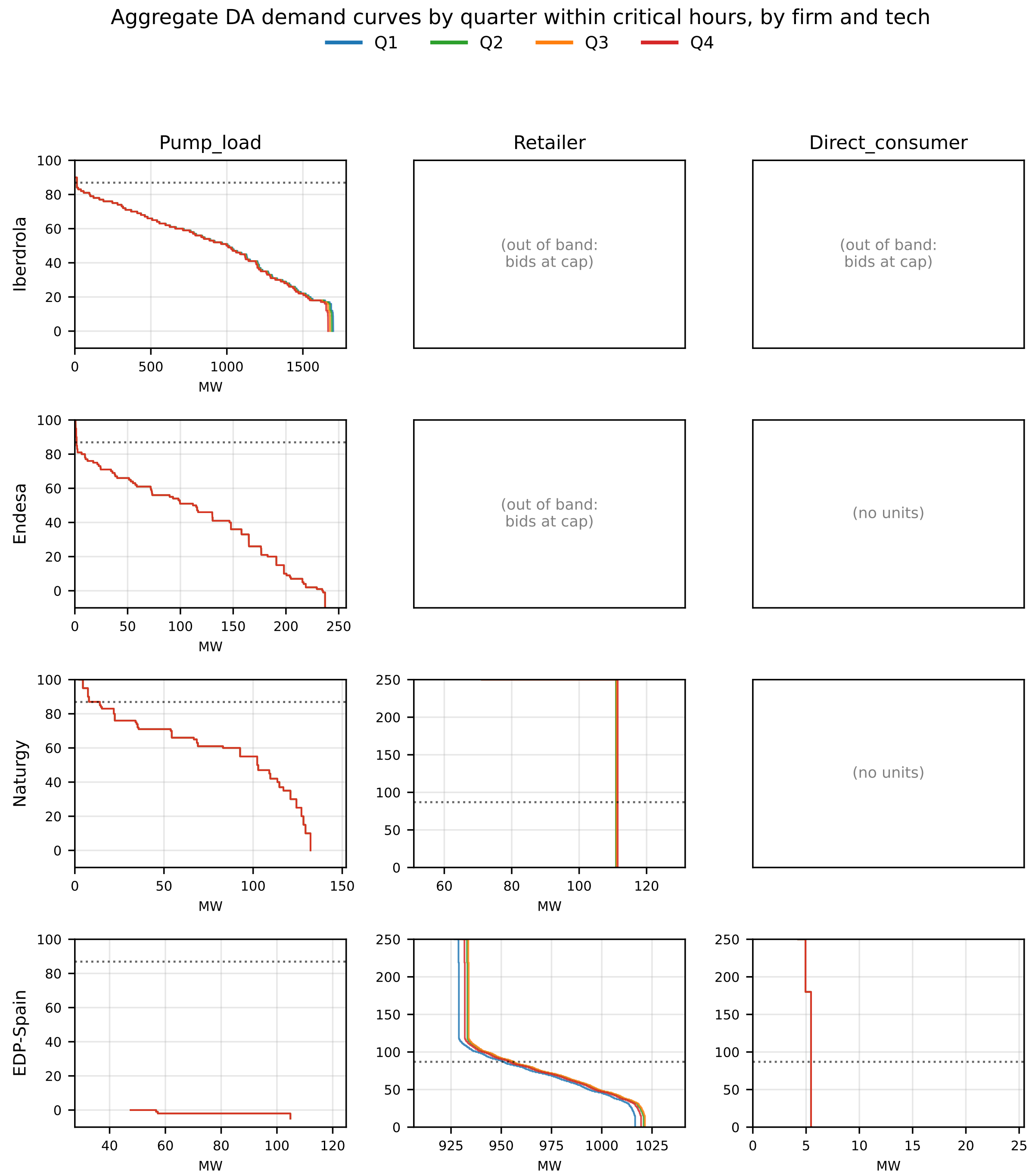

Figure 11: Aggregate DA demand curves by quarter within critical hours, October–December 2025, by firm and demand-side tech-group.

Reading (**mixed**). At the clock-hour level, `Pump_load` shows clear hour-class discrimination for IB, GE and GN (midday-green curves shifted right of flat-blue, both right of critical-red). The Retailer column is empty for IB, GE and Naturgy because their retail buy bids are placed entirely at the price cap — they treat retail as a quantity-securing instrument, not a price signal; only EDP-Spain bids strategically. `Direct_consumer` books are small and concentrated near the cap. At the quarter-within-hour level, every (firm × tech) cell collapses to a single curve: Q1–Q4 overlap. Within-hour granularity is unused on the demand side.

A.13 Per-firm storage-discharge supply curves (DA)

Sell bids only (upward-sloping supply curves). Columns: `Hydro_pump` (pumped-storage discharge) and `Hybrid_RES_storage`.

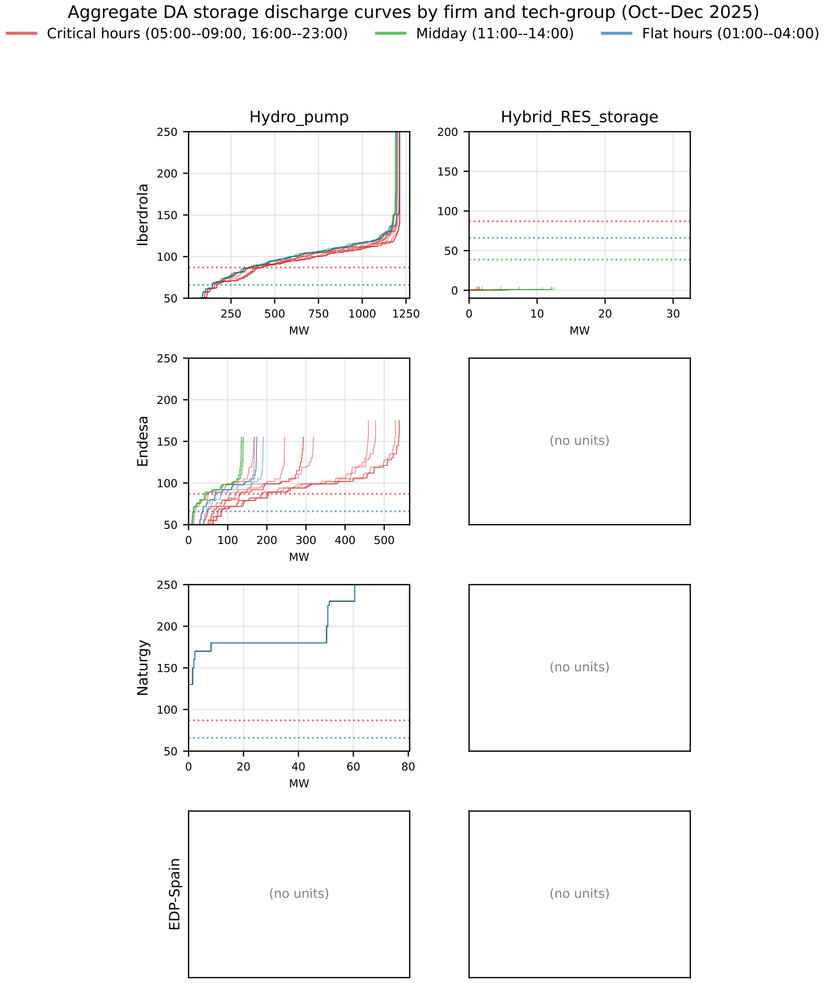

Figure 12: Aggregate DA storage-discharge supply curves by firm and tech-group, October–December 2025, by hour-class. Rows: pivotal firms. Columns: sell-side storage tech-groups.

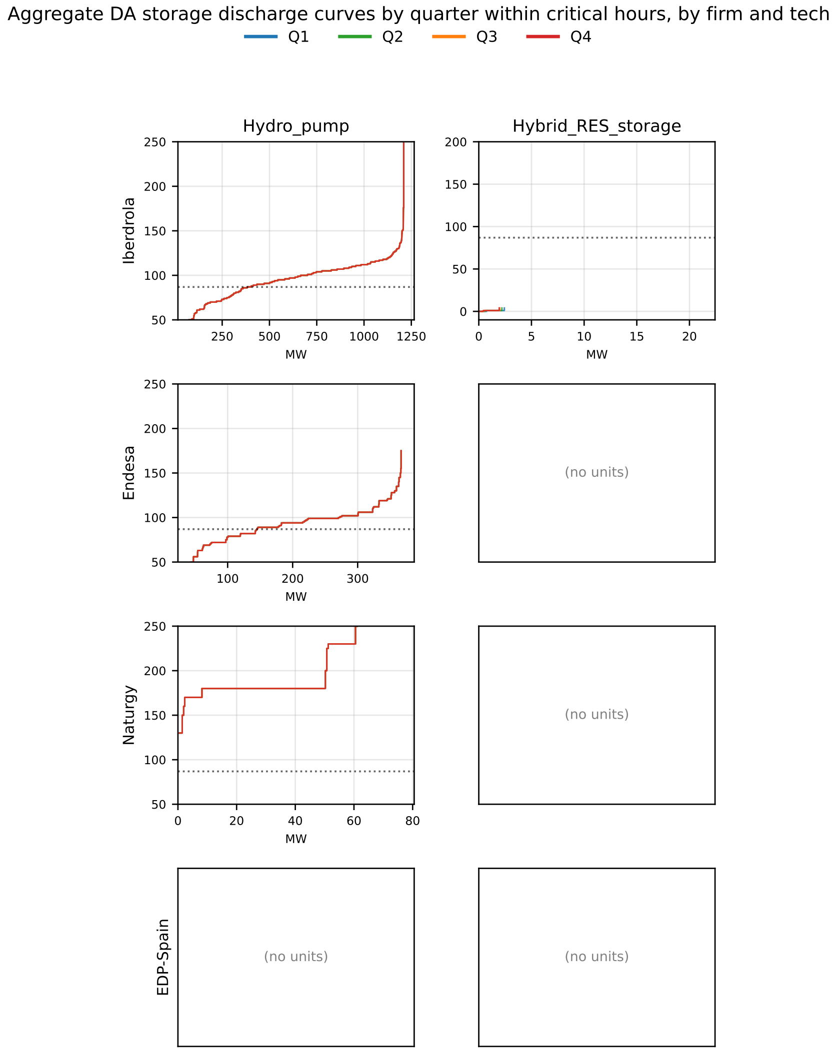

Figure 13: Aggregate DA storage-discharge supply curves by quarter within critical hours, October–December 2025, by firm and tech-group.

Reading (**mixed**). `Hydro_pump` discharge curves separate by hour-class for Iberdrola and Endesa (consistent with discharging more in critical hours when prices are highest); Naturgy is small and concentrated; EDP-Spain has no `Hydro_pump`. `Hybrid_RES_storage` sell-side books are very small (IB-only with meaningful volume) and bid near zero EUR/MWh. As on the demand side, within-hour quarter dispersion is absent across all (firm × tech) cells.

A.14 Per-firm demand-side bid curves, all technologies (IDA)

IDA mirror of Appendix A.12. Same construction on ICAB+IDET data, pooled across the 3 IDA sessions per day.

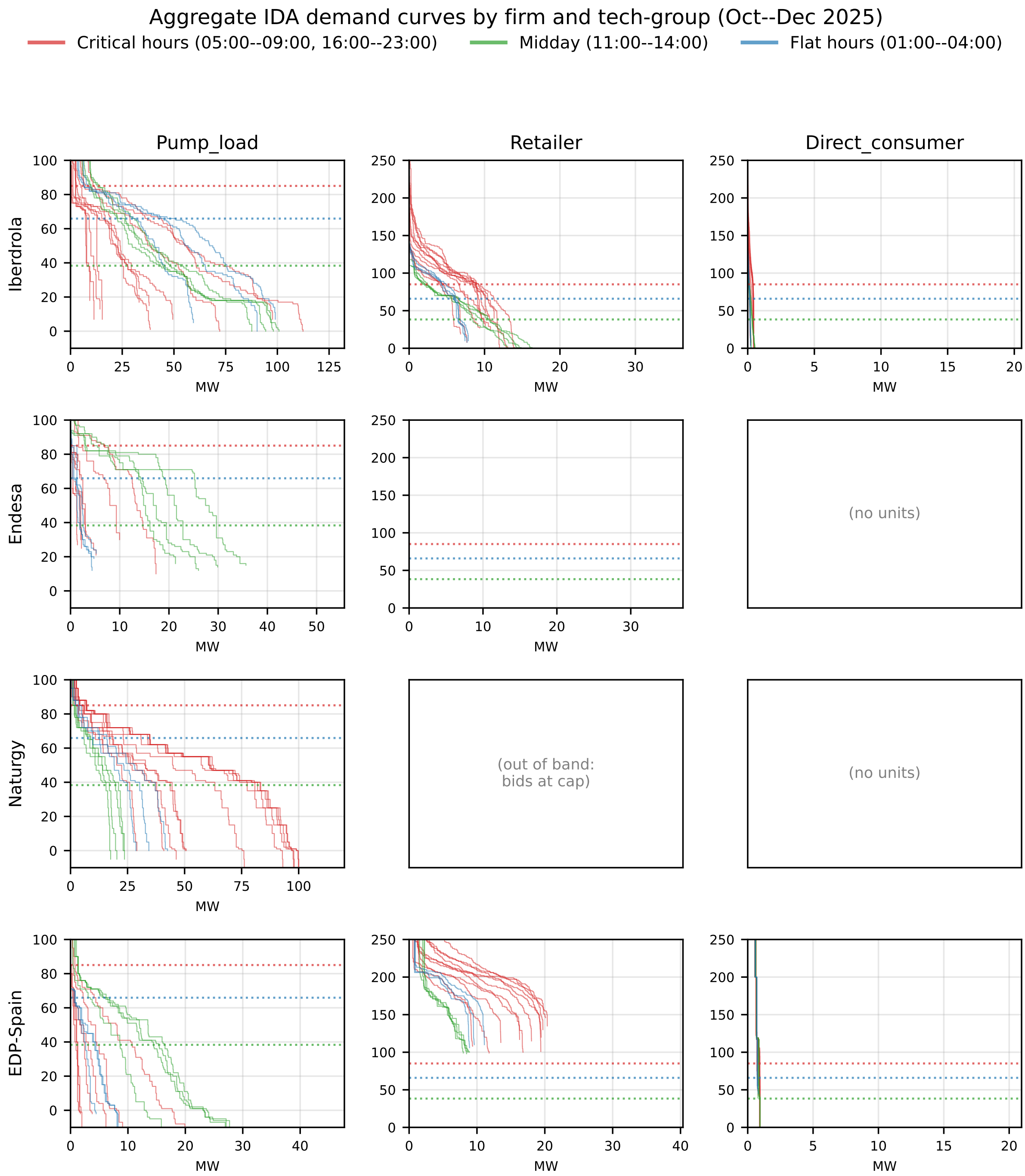

Figure 14: Aggregate IDA demand curves by firm and demand-side tech-group, October–December 2025, by hour-class.

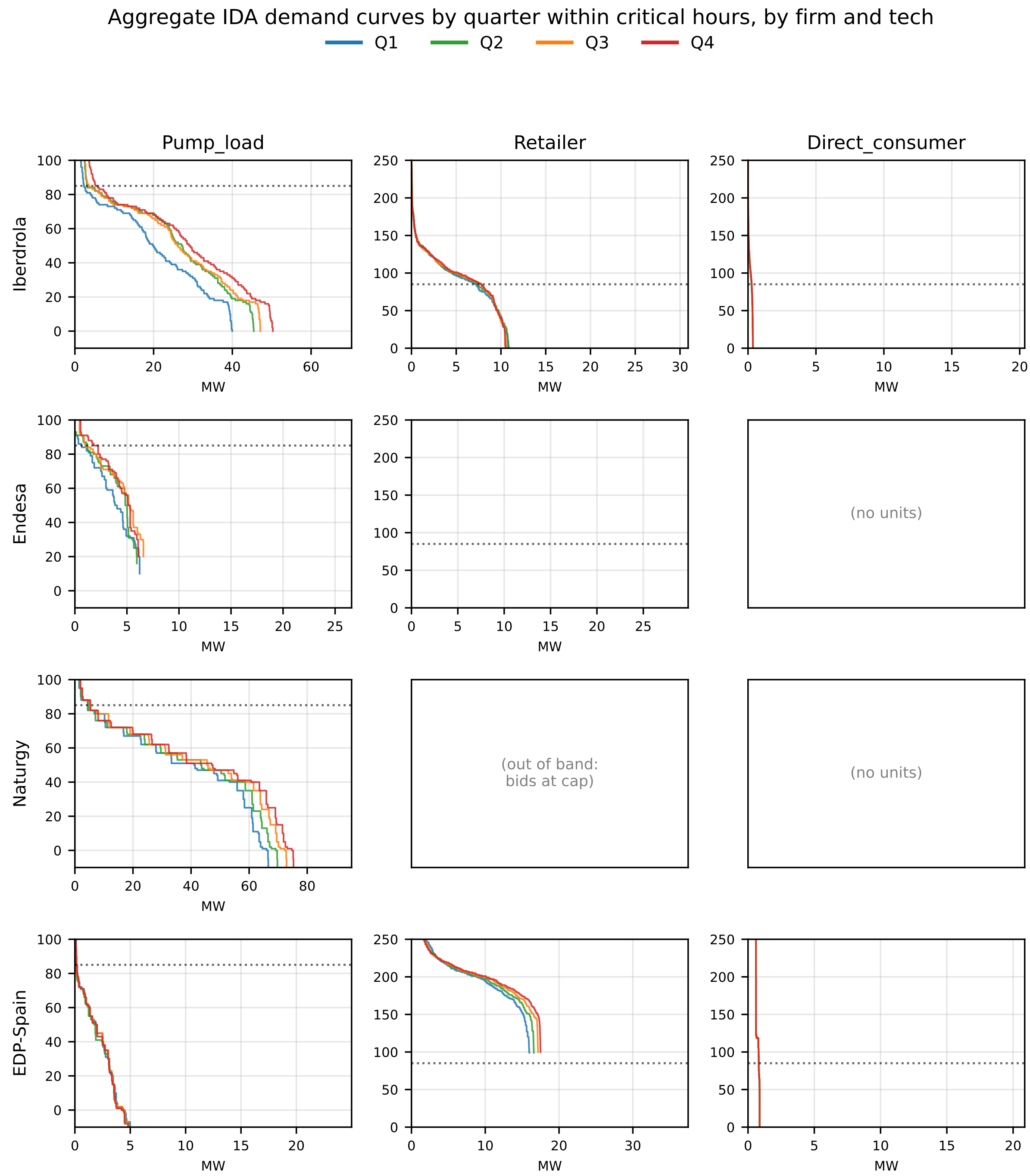

Figure 15: Aggregate IDA demand curves by quarter within critical hours, October–December 2025, by firm and demand-side tech-group.

A.15 Per-firm storage-discharge supply curves (IDA)

IDA mirror of Appendix A.13.

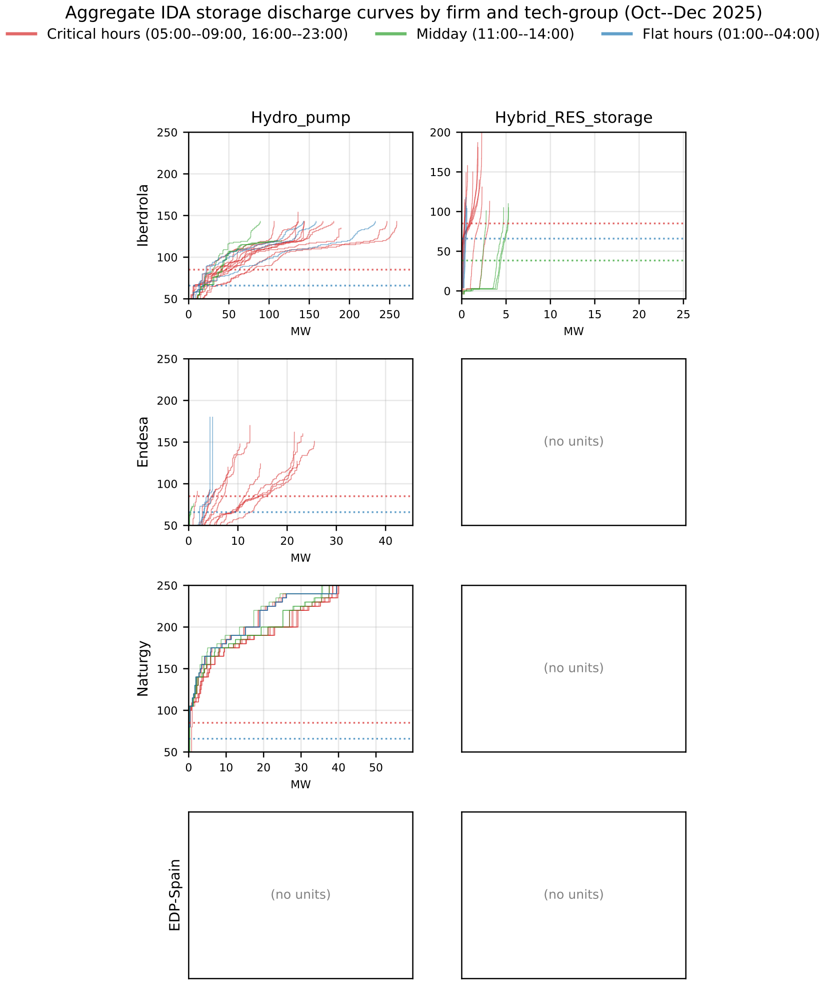

Figure 16: Aggregate IDA storage-discharge supply curves by firm and tech-group, October–December 2025, by hour-class.

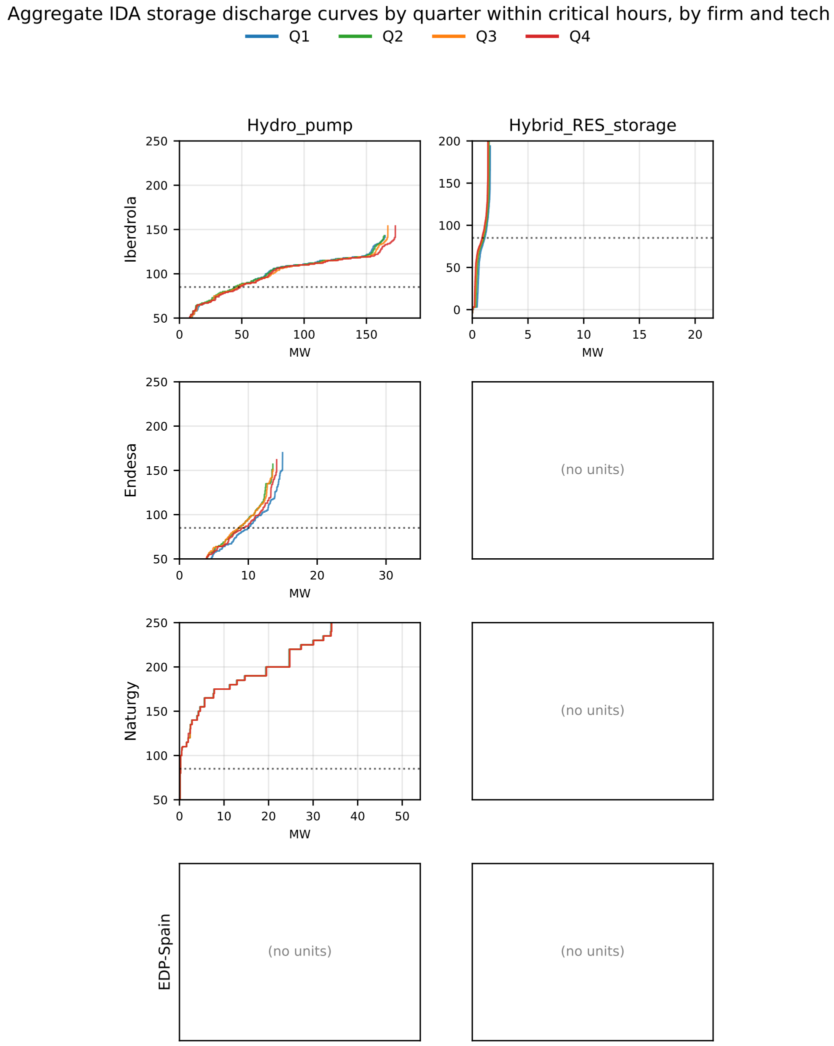

Figure 17: Aggregate IDA storage-discharge supply curves by quarter within critical hours, October–December 2025, by firm and tech-group.

A.16 Within-hour quarter dissimilarity test

Falsification test for the aggregated Q1–Q4 plots: a perfect overlap in the cross-cell average could mask cell-level variation if its direction is idiosyncratic. The diagnostic operates at the (firm, unit, date, hour) cell level, before any aggregation.

Construction. Each quarter's bid in a cell is a discrete measure on the price axis: each tranche (p_k, q_k) contributes a point mass q_k MWh at price p_k EUR/MWh, after integer-rounding the price. Let μ_i denote the bid measure for quarter $i \in \{1, 2, 3, 4\}$, with cumulative distribution

$$F_i(p) \;=\; \sum_{k:\, p_k \le p} q_k,$$

which is exactly the bid-curve step function plotted in §4.2: at price p on the y-axis, $F_i(p)$ is the cumulative MWh offered up to that price.

Dissimilarity between two quarters. For an ordered pair (i, j), we compute the L1 area between the cumulative bid curves over the union of all bid prices present in either quarter:

$$W_1(\mu_i, \mu_j) \;=\; \int_{p_{\min}}^{p_{\max}} \big|F_i(p) - F_j(p)\big|\, dp \;=\; \sum_{k=1}^{K-1} \big|F_i(p_{(k)}) - F_j(p_{(k)})\big| \cdot \big(p_{(k+1)} - p_{(k)}\big), \tag{4}$$

where $\{p_{(1)} < p_{(2)} < \cdots < p_{(K)}\}$ is the sorted union of distinct bid prices from both quarters and $[p_{\min}, p_{\max}] = [p_{(1)}, p_{(K)}]$ is the resulting integration domain. Computation handles the union grid in $O(K \log K)$ via a sort/merge of the two ordered price lists.

Wasserstein interpretation. Equation 4 is the closed-form expression of the 1-D Wasserstein-1 distance (Earth Mover's Distance) for probability measures on $\mathbb{R}$: $W_1(\mu, \nu) = \int |F_\mu - F_\nu|$ (Vallender, 1973). Strictly, the equality requires the two measures to share total mass. In our setting, two quarters may submit slightly different total volumes ($\sum_k q_{i,k} \neq \sum_k q_{j,k}$), in which case Equation 4 is the natural *unbalanced* extension — the L1 area between two non-decreasing step functions over the observed price range. This still captures (i) horizontal shifts of the curve (price re-pricing), (ii) vertical shifts (quantity re-allocation), and (iii) total-volume changes, in a single non-negative number. Magnitudes (in MWh·EUR/MWh) are not directly interpretable, but the metric serves the role we need: $W_1 = 0$ **iff the two bid multisets are identical**, and $W_1 > 0$ iff there is any difference.

Per-cell statistic and flag. For each cell with all four quarters present we compute the maximum pairwise dissimilarity:

$$D \;=\; \max_{1 \le i < j \le 4} W_1(\mu_i, \mu_j).$$

A cell is "flagged" if $D > 10^{-6}$ MWh·EUR/MWh — equivalent to "the four quarters are not bit-identical".

Table 7: Quarter-by-quarter bid dissimilarity, October–December 2025 — three-panel sample comparison. D = max pairwise L1 area between quarter bid curves (Wasserstein-style); D_w = Epanechnikov-weighted (h = 50 EUR/MWh) on the strategic band around the cell's hour-average DA clearing price; ratio = D_w/D on flagged cells. Three sample versions: *Full sample* (all bidding cells); *Strict PS* (cells where the unit was the rank-1 price-setter in ≥ 1 of the 4 quarters of the hour); *Frequent PS* (cells whose unit price-sets in $\geq 1\%$ of all (date, period) cells in the window). Last column flags within-hour heterogeneity in the clearing price (std > 5 EUR/MWh).

Panel	Side	Tech	Firm	Hour-class	N cells	% flagged	median D	median D_w	ratio
	Supply	CCGT	IB	Critical	7,831	5.8	20909.20	0.00	0.00
			IB	Flat	2,139	3.9	2865.34	0.00	0.00
			GE	Critical	5,195	1.8	1897.23	1315.38	0.81
			GE	Flat	1,413	0	–	–	–
			GN	Critical	14,930	24.5	137174.84	4874.85	0.05
			GN	Flat	4,065	0.05	21745.80	1143.33	0.03
			HC	Critical	2,013	0.05	131300.10	2821.70	0.02
			HC	Flat	549	0.55	390.00	0.00	0.00
		Hydro	IB	Critical	6,072	2.1	38.00	6.67	0.03
			IB	Flat	1,656	0.30	39778.46	7040.00	0.18
			GE	Critical	5,552	56.4	1.90	1.60	1.00
			GE	Flat	1,488	47.3	1.60	1.40	1.00
			GN	Critical	2,993	0.03	16.10	5.89	0.37
			GN	Flat	810	0.12	16.00	6.67	0.42

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Panel	Side	Tech	Firm	Hour-class	N cells	% flagged	median D	median D_w	ratio
			HC	Critical	847	0	–	–	–
			HC	Flat	231	0	–	–	–
		Hydro_pump	IB	Critical	1,001	0.40	6158.34	6.67	0.18
			IB	Flat	273	0	–	–	–
			GE	Critical	2,670	0.04	1956.50	1350.73	0.69
			GE	Flat	372	0	–	–	–
			GN	Critical	1,006	0	–	–	–
			GN	Flat	276	0	–	–	–
		Nuclear	IB	Critical	2,907	31.9	205.70	205.70	1.00
			IB	Flat	800	31.6	119.80	119.80	1.00
			GE	Critical	2,473	4.0	67477.50	1500.00	0.02
			GE	Flat	672	1.9	2885.00	486.67	0.84
		Wind	IB	Critical	4,539	97.2	1.10	0.90	1.00
			IB	Flat	1,283	97.3	1.00	0.90	1.00
			GN	Critical	4,152	92.7	0.20	0.00	0.00
			GN	Flat	1,159	92.6	0.13	0.00	0.00
			HC	Critical	3,990	93.7	3.10	0.00	0.00
			HC	Flat	1,081	93.8	3.10	0.00	0.00
		Solar PV	IB	Critical	960	99.5	7.90	7.90	1.00
			GE	Critical	64	3.1	0.10	0.10	1.00
			GN	Critical	1,594	15.9	12.40	12.40	1.00
			GN	Flat	411	0	–	–	–
			HC	Critical	325	100.0	42.12	0.00	0.00
	Demand	Pump_load	IB	Critical	2,558	3.2	480.75	337.89	0.91
			IB	Flat	866	5.5	222.20	217.50	0.97
			GE	Critical	1,370	0	–	–	–
			GE	Flat	617	0	–	–	–
			GN	Critical	1,010	0	–	–	–
			GN	Flat	276	0	–	–	–
			HC	Critical	946	0	–	–	–
			HC	Flat	261	0	–	–	–
		Retailer	IB	Critical	3,423	40.9	33.60	33.60	1.00
			IB	Flat	923	40.2	22.40	22.40	1.00
			GE	Critical	3,971	76.5	26.40	26.40	1.00
			GE	Flat	1,083	76.3	22.65	22.65	1.00
			GN	Critical	4,048	22.0	0.20	0.20	1.00
			GN	Flat	1,104	15.8	0.10	0.10	1.00
			HC	Critical	1,100	0.45	2.24	0.00	0.00
			HC	Flat	300	0	–	–	–
		Direct_consumer	IB	Critical	1,012	95.9	0.30	0.30	1.00
			IB	Flat	276	95.3	0.30	0.30	1.00
			HC	Critical	1,612	15.6	0.10	0.10	1.00
			HC	Flat	441	12.2	0.10	0.10	1.00
	Supply	CCGT	IB	Critical	58	0	–	–	–
			IB	Flat	9	0	–	–	–
			GE	Critical	93	0	–	–	–
			GE	Flat	19	0	–	–	–
			GN	Critical	25	32.0	117685.80	3702.65	0.03
			GN	Flat	4	0	–	–	–
			HC	Critical	29	0	–	–	–
			HC	Flat	4	0	–	–	–

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Panel	Side	Tech	Firm	Hour-class	N cells	% flagged	median D	median D_w	ratio
		Hydro	IB	Critical	408	1.5	5075.55	189.78	0.08
			IB	Flat	60	0	–	–	–
			GE	Critical	29	58.6	130.84	52.57	0.38
			GE	Flat	10	30.0	97.29	36.51	0.36
			GN	Critical	58	0	–	–	–
			GN	Flat	4	0	–	–	–
			HC	Critical	15	0	–	–	–
		Hydro_pump	IB	Critical	246	0.41	12278.65	11927.95	0.97
			IB	Flat	13	0	–	–	–
			GE	Critical	113	0	–	–	–
			GE	Flat	13	0	–	–	–
			GN	Critical	1	0	–	–	–
		Nuclear	GE	Critical	11	0	–	–	–
			GE	Flat	1	0	–	–	–
		Wind	IB	Critical	12	100.0	16.18	16.15	1.00
			IB	Flat	3	100.0	9.00	9.00	1.00
			GN	Critical	3	100.0	0.00	0.00	0.99
			GN	Flat	1	100.0	0.30	0.30	1.00
			HC	Critical	1	100.0	5.72	5.70	1.00
			HC	Flat	1	100.0	22.36	22.28	1.00
		Solar PV	IB	Critical	3	100.0	9.00	9.00	1.00
			GN	Critical	2	100.0	43.80	43.80	1.00
			HC	Critical	1	100.0	4.16	4.14	1.00
	Supply	CCGT	GE	Critical	1,958	0.15	11704.00	2887.94	0.25
			GE	Flat	534	0	–	–	–
			HC	Critical	2,013	0.05	131300.10	2821.70	0.02
			HC	Flat	549	0.55	390.00	0.00	0.00
		Hydro	IB	Critical	4,048	2.4	115.31	6.67	0.12
			IB	Flat	1,104	0.27	41059.21	7266.67	0.18
			GE	Critical	1,679	77.7	32.45	4.40	0.52
			GE	Flat	477	64.2	2.30	1.60	0.53
			GN	Critical	1,012	0	–	–	–
			GN	Flat	276	0.36	16.00	6.67	0.42
		Hydro_pump	IB	Critical	1,001	0.40	6158.34	6.67	0.18
			IB	Flat	273	0	–	–	–
			GE	Critical	1,979	0.05	1956.50	1350.73	0.69
			GE	Flat	320	0	–	–	–
		Wind	IB	Critical	468	98.9	2.70	0.00	0.00
			IB	Flat	131	98.5	2.48	0.00	0.00
			GN	Critical	3,027	96.2	0.20	0.00	0.39
			GN	Flat	826	96.7	0.06	0.00	0.00
			HC	Critical	3,990	93.7	3.10	0.00	0.00
			HC	Flat	1,081	93.8	3.10	0.00	0.00
		Solar PV	IB	Critical	808	99.9	8.20	8.20	1.00
			GE	Critical	23	0	–	–	–
			GN	Critical	99	26.3	0.10	0.10	1.00
			GN	Flat	12	0	–	–	–
			HC	Critical	325	100.0	42.12	0.00	0.00

Reading (**mixed**). Pump-storage is genuinely mechanical: `Pump_load` and `Hydro_pump` flag $\leq$ 5.5% across all firms and hour-classes, confirming the within-hour null in §4.3. Naturgy

CCGT flags 24.5% in critical vs 0.0% in flat, matching the visual Q1–Q4 separation seen in `provisional.tex` §2. By contrast `Wind` and `Solar PV` flag $\geq 92\%$ for IB, GN and HC — bids vary across the four quarters in almost every cell, though for Wind the magnitudes are small (median $D \leq 2.6$), consistent with quarter-by-quarter automated forecast updates rather than strategic re-pricing. `Retailer` flags 41%, 76% and 22% for IB, GE and GN with Endesa's median $D = 11.2$: pivotal firms do vary their retail demand bids across quarters in ways the aggregated demand-curve plot of §A.12 does not display. Endesa Hydro stands out on the supply side with 56% / 47% flag rates.

Kernel-weighted dissimilarity (D_w). A second column D_w in Table 7 reports the same L1 area under an Epanechnikov-kernel weight centred at the period's mean DA clearing price, with bandwidth $h = 20$ EUR/MWh:

$$D_w = \max_{i<j} \int \left|F_i(p) - F_j(p)\right| \cdot K_h(p - \bar{p}_{\text{clear}})\, dp, \qquad K_h(u) = \tfrac{3}{4h}\left(1 - (u/h)^2\right)_+.$$

The kernel has compact support on $[\bar{p}_{\text{clear}} - h, \bar{p}_{\text{clear}} + h]$, so D_w only counts bid differences inside the "strategically-relevant" band where bid-shading plausibly affects clearing. Mass differences at the price cap (4,000 EUR/MWh) contribute zero to D_w but contribute to the unweighted D. Comparing the two columns sharpens interpretation: a high D with a low ratio D_w/D means the within-quarter variation lives at the extremes (price-cap padding, not strategic); a high D with a ratio near 1 means the variation is concentrated around clearing (strategic re-pricing).

A.17 Price-setting unit on illustrative days

The aggregate quarter-curves in §4.2 average over ∼92 dates and so smooth daily shocks. The kernel-weighted D_w partially addresses this by operating at the (firm, unit, date, hour) cell, but it still aggregates across days for its summary statistics. As a visual complement, this subsection plots the *price-setting unit's* full bid ladder per quarter for two single days: one high-demand winter weekday and one low-demand flat Sunday.

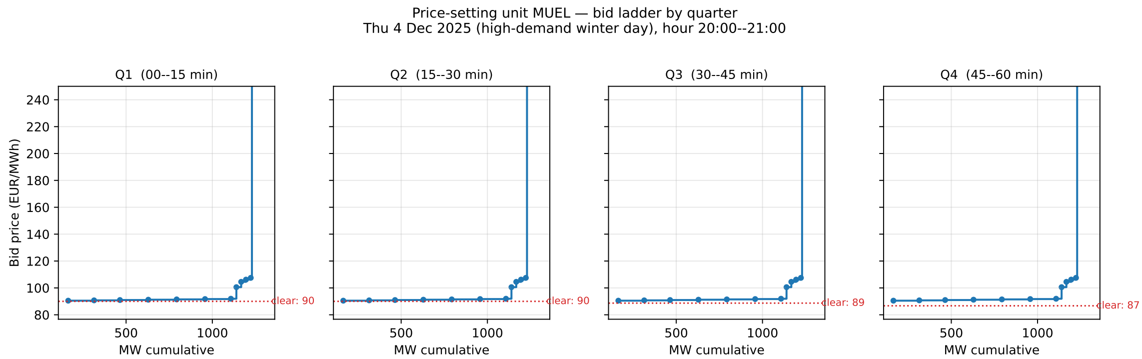

Figure 18: Price-setting unit on a high-demand winter day (Thu 4 Dec 2025, hour 20:00–21:00). The unit identified as the price-setter (highest accepted sell tranche within 0.5 EUR/MWh of the period's DA clearing price) is MUEL — a Naturgy CCGT block — in two of the four quarters. Each panel shows the unit's full bid ladder for that quarter; the red dotted line marks the period's clearing price.

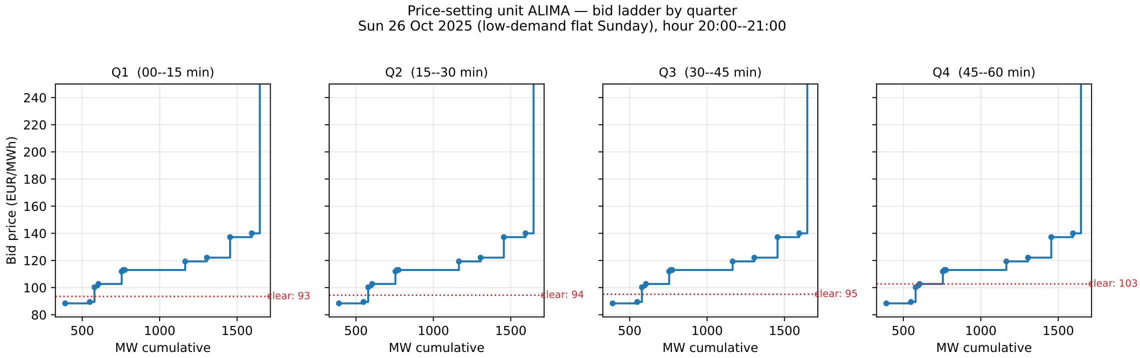

Figure 19: Price-setting unit on a flat low-demand Sunday (Sun 26 Oct 2025, hour 20:00–21:00). The same construction. Even on a flat day the four price-setting tranches differ visibly across quarters — but the unit identity itself rotates more freely (no concentrated quarter-by-quarter shaping by one focal unit).

Reading (**supportive**). On the high-demand day, the same Naturgy CCGT (MUEL) sets the price in two adjacent quarters with a small upward step between them — the kind of micro-arbitrage the within-day model predicts. On the flat Sunday the price-setting unit itself rotates across the four quarters, and the bid ladders look more like four independent units bidding around a similar marginal cost rather than one unit shaping its ladder. These two examples illustrate the per-cell variation that the aggregate plots compress into a single smoothed line. A systematic per-day quarter-dissimilarity sweep is the next natural extension; we leave it for the appendix of a future version.

A.18 Pre-trend diagnostic

The parallel-trends assumption requires $\mathbb{E}[Y_t(0) - Y_0(0) \mid \text{crit}] = \mathbb{E}[Y_t(0) - Y_0(0) \mid \text{flat}]$ in the pre-reform window: the time-evolution of the counterfactual outcome should be the same for critical and flat hours. The visual diagnostic below plots, for each outcome, the cumulative deviation of the monthly mean from its January-2024 baseline in critical hours (red) and flat hours (blue). Two lines that overlap pre-reform support the assumption.

A.18.1 Outcome 1: q_2 (intraday upward repositioning)

OJO, MOEVE SHOWS A HUGE CHANGE IN BEHAVIOUR AFTER THE ID15 REFORM!! THE REST SEEM NICE!!

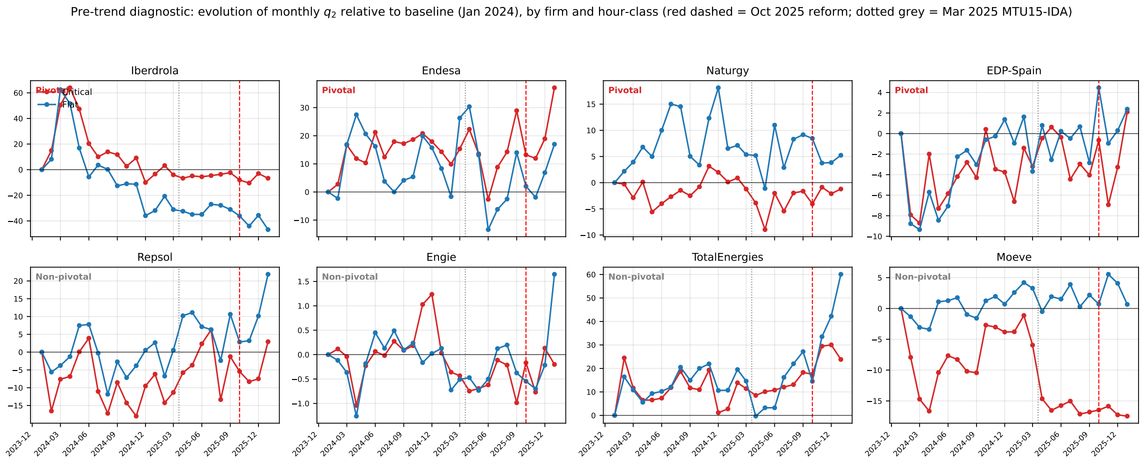

Figure 20: Per-firm evolution of monthly q_2 relative to Jan-2024 baseline, by hour-class, Jan 2024 – Dec 2025. Each panel plots, for one firm, the cumulative deviation of monthly mean q_2 from its January-2024 value in critical hours (red) and flat hours (blue). Parallel-trends test: the two lines should track each other pre-reform. Red dashed: MTU15-DA reform. Grey dotted: MTU15-IDA. EDP-PT omitted (no PIBCI entries).

OJO, ID15 IS VERY IMPORTANT FOR WIND!! I SHOULD THINK ABOUT HOW IT AFFECTS!!

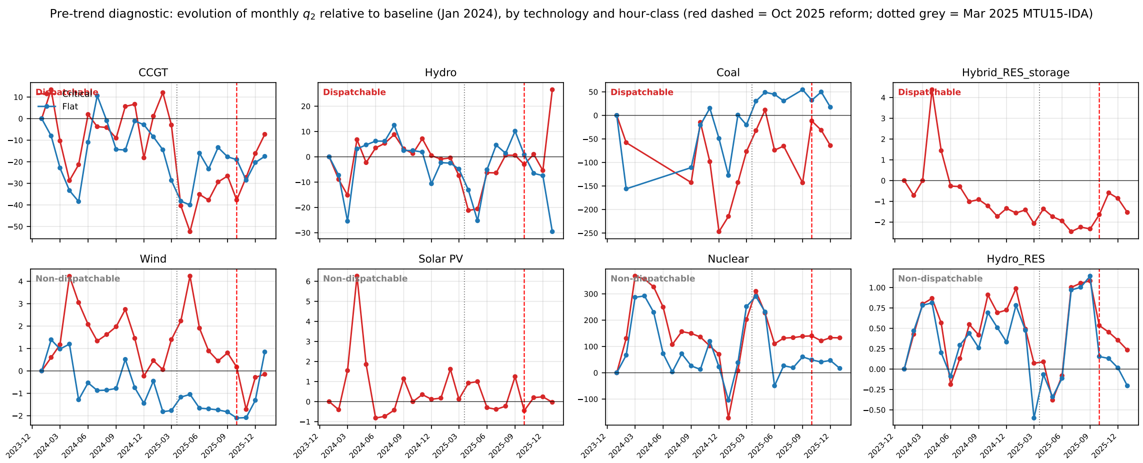

Figure 21: Per-technology evolution of monthly q_2 relative to Jan-2024 baseline, by hour-class, Jan 2024 – Dec 2025. Same construction as Figure 20 but grouped by `tech_group` instead of parent. Sample restricted to pivotal + non-pivotal firms. Solar PV has no flat-hour observations by construction (no generation at 1–3 AM); Coal sample is small.

Reading (**mixed**). For most firms the red and blue lines track each other pre-reform, suggesting parallel trends are plausible at the firm level. The exception is Moeve, whose flat-hour series shifts down sharply at MTU15-IDA (Mar 2025) and stays below baseline thereafter, while the critical-hour series stays near zero — a structural break in flat-hour behaviour that is concentrated in this single non-pivotal firm. By technology, CCGT and Hydro pre-trends look reasonable; Wind, Nuclear and Hydro-RES also co-move pre-reform. The headline $\beta_3 \approx +3.87$ sits inside the month-to-month noise band, so the DiD identification leans on averaging across day-clusters within the same-cal-month window rather than on any single month.

A.18.2 Outcome 2: clearing prices (IDA and DA)

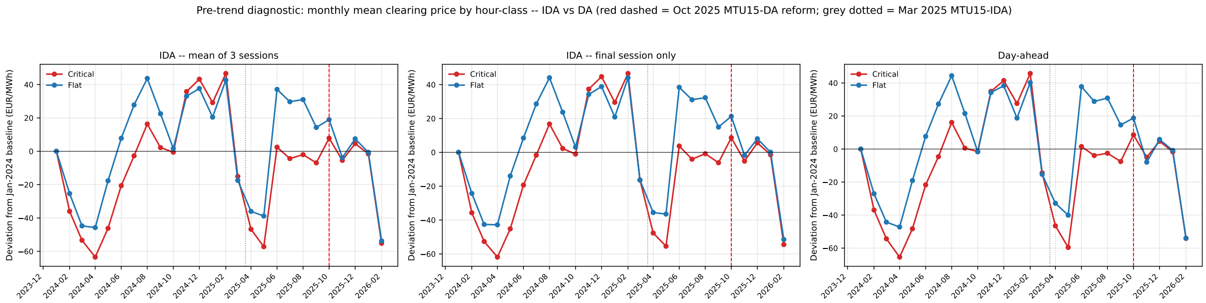

Figure 22: Evolution of monthly mean Spanish clearing price relative to Jan-2024 baseline, by hour-class, Jan 2024 – Dec 2025. Left: arithmetic mean across the three IDA auction sessions covering each (date, clock-hour). Middle: only the last (final) IDA session covering each slot. Right: day-ahead clearing price. All three average across the MTU15 periods that map to the same clock-hour before averaging within (month, hour-class). Red dashed: MTU15-DA reform. Grey dotted: MTU15-IDA.

Reading (**supportive**). All three markets tell the same story. Pre-MTU15-IDA, the critical and flat lines track each other tightly through the seasonal cycle (down to −60 EUR/MWh in spring 2024, up to +45 by Feb 2025): the crit-flat price spread is stable, parallel trends look strong. After MTU15-IDA (Mar 2025) the lines diverge: flat-hour prices rise +30 to +40 above baseline while critical-hour prices stay near zero, compressing the peak-flat spread by 30–40 EUR/MWh on average. This is consistent with the strategic-granularity story — the reform lets dispatchable units re-price into critical hours, pulling those clearing prices down relative to flat hours where there is no analogous opportunity. The fact that the pattern is essentially identical in DA and IDA reflects the arbitrage-linked nature of the sequential markets: the DA-to-IDA wedge is small (Appendix A.6). The mean-of-3-sessions and final-session-only IDA variants are visually indistinguishable.

A.19 Time placebos

A.20 Triple-difference: pivotality as moderator

$$\begin{aligned} q_{2,fudh} = \alpha &+ \beta_1\,\text{crit}_h + \beta_2\,\text{post}_d + \beta_3\,\text{piv}_f \\ &+ \beta_{12}\,\text{crit}_h\,\text{post}_d + \beta_{13}\,\text{crit}_h\,\text{piv}_f + \beta_{23}\,\text{post}_d\,\text{piv}_f \\ &+ \beta_{123}\,\text{crit}_h\,\text{post}_d\,\text{piv}_f + \gamma_f + \delta_{\text{DOW}(d)} + \boldsymbol{\theta}'X_{dh} + \varepsilon_{fudh}, \end{aligned} \quad (5)$$

where $X_{dh} = (\text{wind}_{dh}, \text{solar}_{dh}, \text{demand}_{dh})$ in GW.

B Data appendix

B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

B.2 Firms in the analysis sample (detailed)

I THINK WE CAN TAKE THE CAPAPCITY DATA! BUT MAYBE IT IS ONLY FOR EXPORTS AND IMPORTS!.

Column definitions. *DA auction share*: PDBF offer-type 1, anonymous DA clearing. *DA bilateral share*: PDBF offer-type 4, pre-negotiated firm-to-firm transfers. *IDA cumulative share*: PIBCA accumulated (last session per unit-period). *Net position*: net seller if 2025 DA sales $> 60\%$ of turnover, net buyer if $< 40\%$, mixed otherwise. The net-position label is descriptive; vertical integration does not enter the identification.

Table 8: Firms in the analysis sample (detailed). Channel-segregated 2025 shares. Channel denominators are channel-specific (they do not sum across rows).

Parent firm	Gen. units	Capacity (GW)	DA auction share (%)	DA bilateral share (%)	IDA cumul. share (%)	Net position
Dominant operators in Spanish generation (CNMC tier)						
Iberdrola	70	31.03	20.01	27.18	20.37	Mixed
Endesa	38	12.46	36.88	7.11	12.64	Net buyer
Naturgy	39	12.60	3.86	16.18	7.67	Net buyer
EDP-Spain	21	4.73	0.77	5.67	2.60	Net buyer
EDP-Portugal	16	7.42	20.62	12.70	5.17	—
Other large operators (non-dominant)						
Repsol	13	0.61	1.79	1.26	0.62	Net buyer
Engie España	59	4.05	0.03	2.49	1.07	Net seller
TotalEnergies	15	1.19	1.19	0.38	0.59	Net buyer
Moeve	112	1.28	0.03	—	0.66	Net buyer

Reading (**descriptive**). Iberdrola is roughly balanced across channels. Endesa is auction-heavy. Naturgy is bilateral-heavy. Vertical-integration status is descriptive only and does not enter the identification.

B.3 File-format specifications and parser conventions

B.4 Power-vs-energy unit discipline

B.5 Coverage and missingness

B.6 Hour-of-day classification metrics

Table 9: Hour-of-day classification metrics, Spain 2025. Mean load, solar, net-load (= load − wind − solar), within-hour SD of net-load, and signed within-hour load change $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$, by clock-hour.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	Δ demand (MW)	Class
00:00–01:00	22.8	0.1	7.3	15.4	412	−834	*dropped*
01:00–02:00	22.0	0.1	7.1	14.8	347	−470	*flat*
02:00–03:00	21.6	0.1	6.9	14.6	336	−190	*flat*
03:00–04:00	21.5	0.1	6.7	14.7	357	176	*flat*
04:00–05:00	22.4	0.1	6.7	15.7	587	1114	*dropped*
05:00–06:00	24.4	0.2	6.7	17.5	721	1743	*critical*
06:00–07:00	26.5	1.5	6.6	18.4	915	1423	*critical*
07:00–08:00	27.7	4.9	6.2	16.6	1078	627	*critical*
08:00–09:00	28.3	9.1	5.8	13.4	1006	253	*critical*
09:00–10:00	28.3	12.0	5.5	10.9	753	−104	*dropped*
10:00–11:00	28.2	13.3	5.4	9.5	499	−66	*dropped*
11:00–12:00	28.2	13.8	5.5	8.9	468	94	*dropped*
12:00–13:00	28.3	13.8	5.7	8.8	502	−104	*dropped*
13:00–14:00	27.9	13.5	5.8	8.5	485	−331	*dropped*
14:00–15:00	27.5	12.9	6.1	8.5	469	−122	*dropped*
15:00–16:00	27.6	11.7	6.4	9.6	677	291	*dropped*
16:00–17:00	28.2	9.0	6.8	12.3	1109	553	*critical*
17:00–18:00	29.2	5.4	7.4	16.4	1303	891	*critical*
18:00–19:00	30.5	2.2	7.7	20.5	1065	884	*critical*
19:00–20:00	31.3	0.6	7.9	22.8	522	176	*critical*
20:00–21:00	30.4	0.3	7.9	22.2	622	−1444	*critical*
21:00–22:00	28.0	0.2	7.9	20.0	781	−1851	*critical*
22:00–23:00	25.8	0.2	7.7	18.0	664	−1407	*critical*
23:00–00:00	24.2	0.1	7.5	16.5	510	−1062	*dropped*

Reading (**supportive**). Critical hours (05:00–09:00, 16:00–23:00) have within-hour SD of net-load several times higher than flat hours (01:00–04:00). The ranking pins the classification directly to the structural variability of residual demand.

References

Alberto Abadie. Semiparametric difference-in-differences estimators. *The Review of Economic Studies*, 72(1):1–19, 2005.

Blaise Allaz and Jean-Luc Vila. Cournot competition, forward markets and efficiency. *Journal of Economic Theory*, 59(1):1–16, 1993.

James B. Bushnell. A mixed complementarity model of hydro-thermal electricity competition in the western US. *Operations Research*, 51(1):80–93, 2003.

Clément de Chaisemartin and Xavier D'Haultfœ uille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–96, 2020.

James J. Heckman, Hidehiko Ichimura, and Petra E. Todd. Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64(4):605–654, 1997.

Ali Hortaçsu and Steven L. Puller. Understanding strategic bidding in multi-unit auctions: a case study of the texas electricity spot market. *RAND Journal of Economics*, 39(1):86–114, 2008.

Koichiro Ito and Mar Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–57, 2016.

Erin T. Mansur. Measuring welfare in restructured electricity markets. *Review of Economics and Statistics*, 90(2):369–386, 2008.

Pedro H. C. Sant'Anna. Causal inference using difference-in-differences, lecture 5: How covariates can make your did more plausible. Emory University, January 2025, 2025.

Pedro H. C. Sant'Anna and Jun Zhao. Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122, 2020.

S. S. Vallender. Calculation of the Wasserstein distance between probability distributions on the line. *Theory of Probability & Its Applications*, 18(4):784–786, 1973.

Frank A. Wolak. Measuring unilateral market power in wholesale electricity markets: the california market, 1998–2000. *American Economic Review*, 93(2):425–430, 2003.